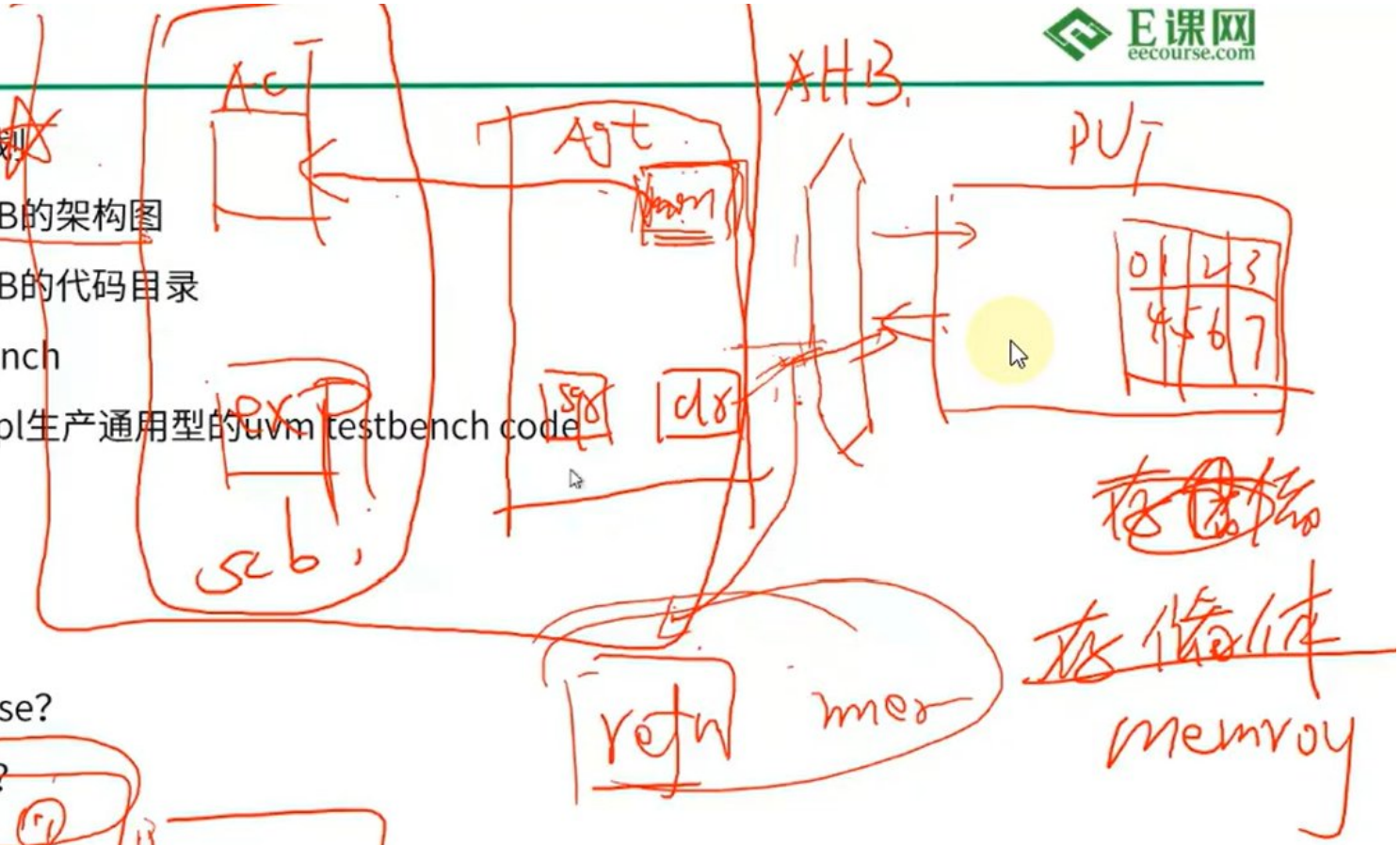

E课网
eecourse.com
摩尔精英
MooreElite.com
半导体集成电路高新技术教育培训课程系列之
数字IC验证工程师就业班
UVM实训项目01 - AHB_SRAMC
第一节 UVM TB的搭建和测试案例的编写
0:03:29 / 1:32:48
护眼: 关
x2倍速
高清

内容



- AHB-SRAMC UVMTB验证计划
- AHB-SRAMC验证平台UVMTB的架构图
- AHB-SRAMC验证平台UVMTB的代码目录
- 如何构建通用的UVM Testbench
- 利用perl脚本 uvm_tb_gen.pl生产通用型的uvm testbench code
- 如何编写interface code?
- 如何编写driver?
- 如何编写sequence item?
- 如何编写sequence & testcase?
- 如何为激励添加constraints?
- 如何建立regression?
- 如何统计和分析code coverage?
- 如何查看testcase passing rate?

0:08:08 / 1:32:48 护眼: 关 x2倍速 高清

内容

E课网 eecourse.com

- AHB-SRAMC UVMTB验证计划
- AHB-SRAMC验证平台UVMTB的架构图
- AHB-SRAMC验证平台UVMTB的代码目录
- 如何构建通用的UVM Testbench
- 利用perl脚本 uvm_tb_gen.pl生产通用型的uvm testbench code
- 如何编写interface code?
- 如何编写driver?
- 如何编写sequence item?
- 如何编写sequence & testcase?
- 如何为激励添加constraints?
- 如何建立regression?
- 如何统计和分析code coverage?
- 如何查看testcase passing rate?

0:11:20 / 1:32:48 护眼: 关 x2倍速 高清

AHB-SRAMC UVMTB的验证计划

2 模块设计 5
2.1 设计架构 5
2.2 接口信号 5
3 模型或者总线功能模型 .....
4 UVM testbench 验证平台 .....
4.1 验证语言：SystemVerilog .....
4.2 验证方法： UVM .....
4.3 验证工具：VCS/Questasim .....
4.4 验证目标： 8
4.4.1 RTL 和 AHB-SRAMC 设计规格一致 8
4.4.2 代码覆盖率 8
4.4.3 功能覆盖率 8
4.5 验证策略： 8
4.5.1 白盒验证：AHB-SRAMC RTL 可见，需要熟悉 RTL 设计实现方法 8
4.5.2 覆盖率驱动的验证 8
4.5.3 带约束的随机化验证策略 9
4.6 人员安排 9
4.7 时间安排 9
5 AHB-SRAMC testcase plan 9
6 测试案例 12
6.1 直接测试案例 12
6.2 随机测试案例 15
7 覆盖率 16
7.1 功能覆盖率 16

详细介绍验证计划

0:12:43 / 1:32:48 护眼: 关 x2倍速 高清

AHB-SRAMC UVMTB的验证计划

E课网 eecourse.com

2 模块设计 5
2.1 设计架构 5
2.2 接口信号 5
3 模型或者总线功能模型 .....
4 UVM testbench 验证平台 .....
4.1 验证语言：SystemVerilog .....
4.2 验证方法： UVM .....
4.3 验证工具：VCS/Questasim .....
4.4 验证目标： 8
4.4.1 RTL 和 AHB-SRAMC 设计规格一致 8
4.4.2 代码覆盖率 8
4.4.3 功能覆盖率 8
4.5 验证策略： 8
4.5.1 白盒验证：AHB-SRAMC RTL 可见，需要熟悉 RTL 设计实现方法 8
4.5.2 覆盖率驱动的验证 8
4.5.3 带约束的随机化验证策略 9
4.6 人员安排 9
4.7 时间安排 9
5 AHB-SRAMC testcase plan 9
6 测试案例 12
6.1 直接测试案例 12
6.2 随机测试案例 15
7 覆盖率 16
7.1 功能覆盖率 16

详细介绍验证计划

① 对spec理解 测试点的

② scenario

③ checker

④ function coverage

www.coreshinelearning.com userhd7y06

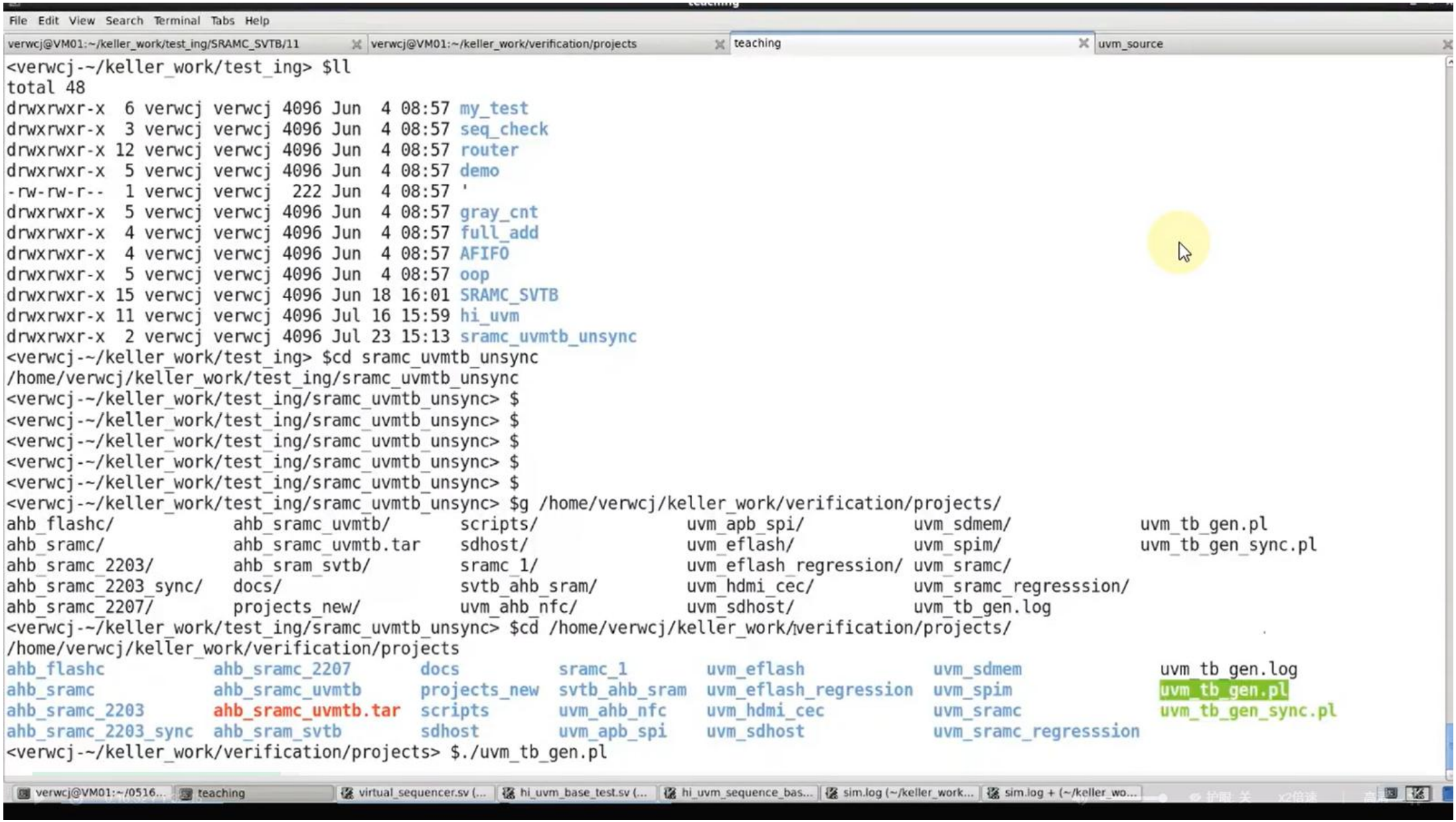
0:12:43 / 1:32:48 护眼: 关 x2倍速 高清

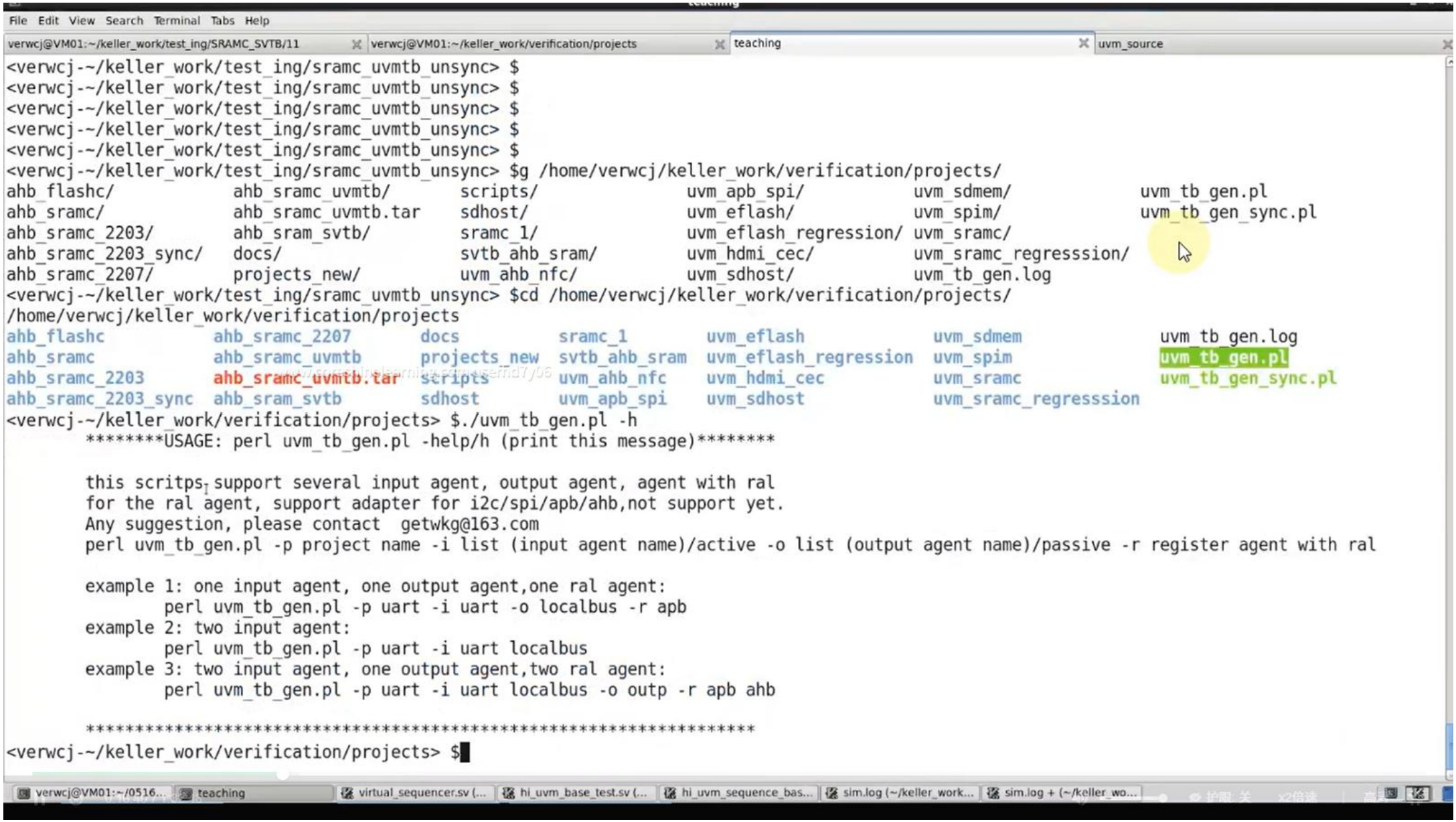
如何构建UVMTB的通用型代码

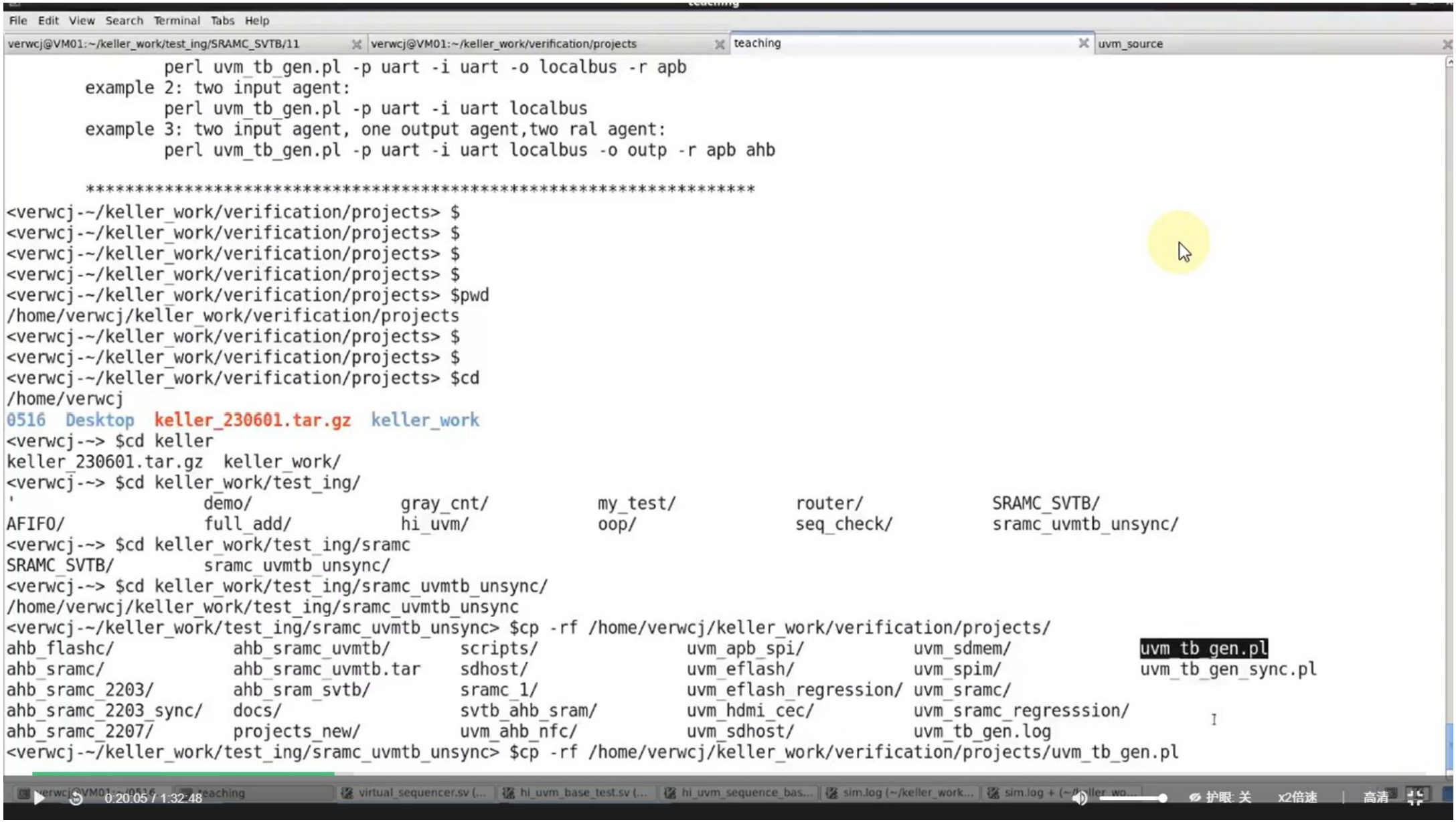
- uvm_tb_gen.pl
- 讲解使用方法： ./uvm_tb_gen.pl -help

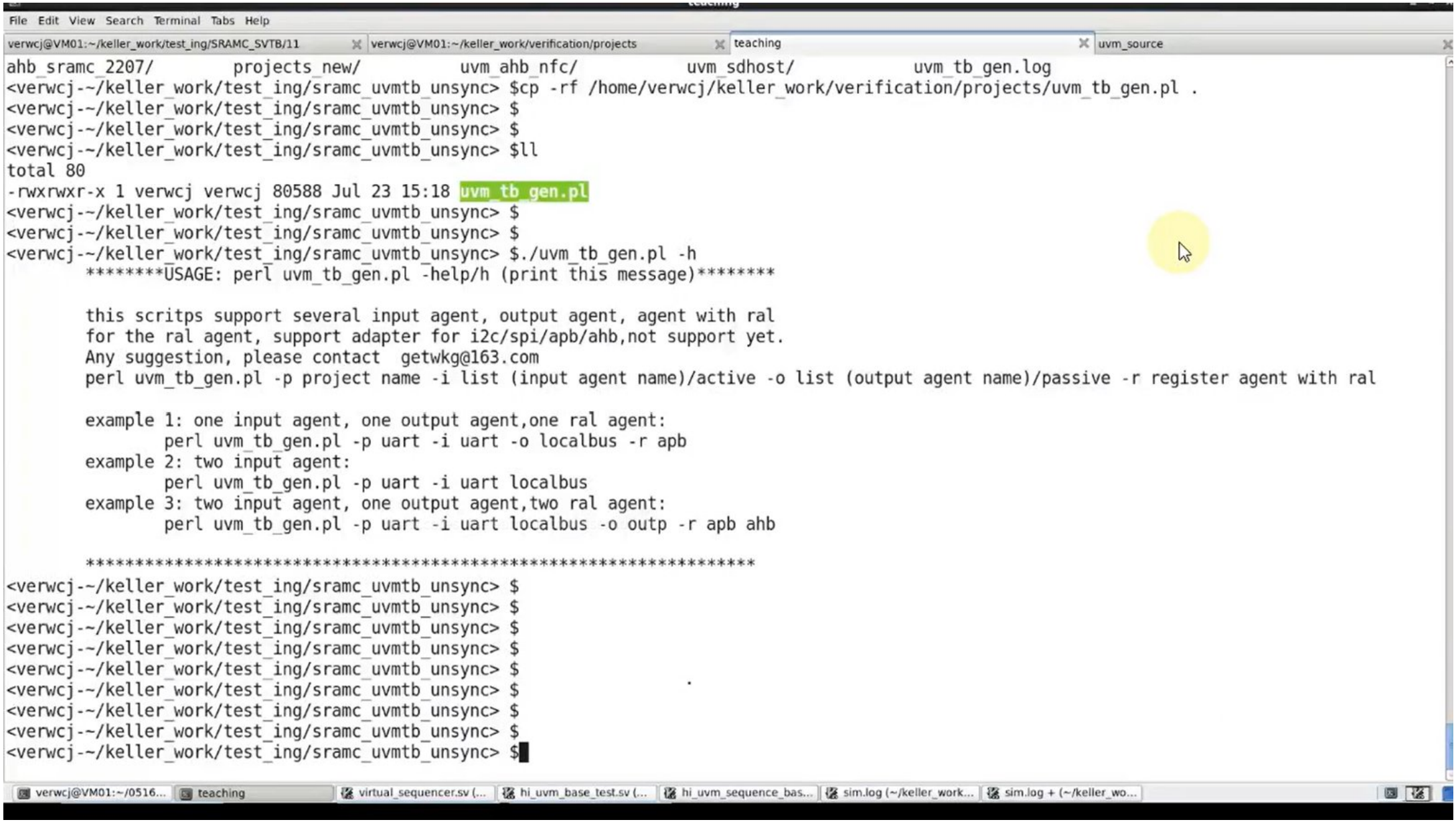
www.cbreshinelearning.com.user/id7/y05

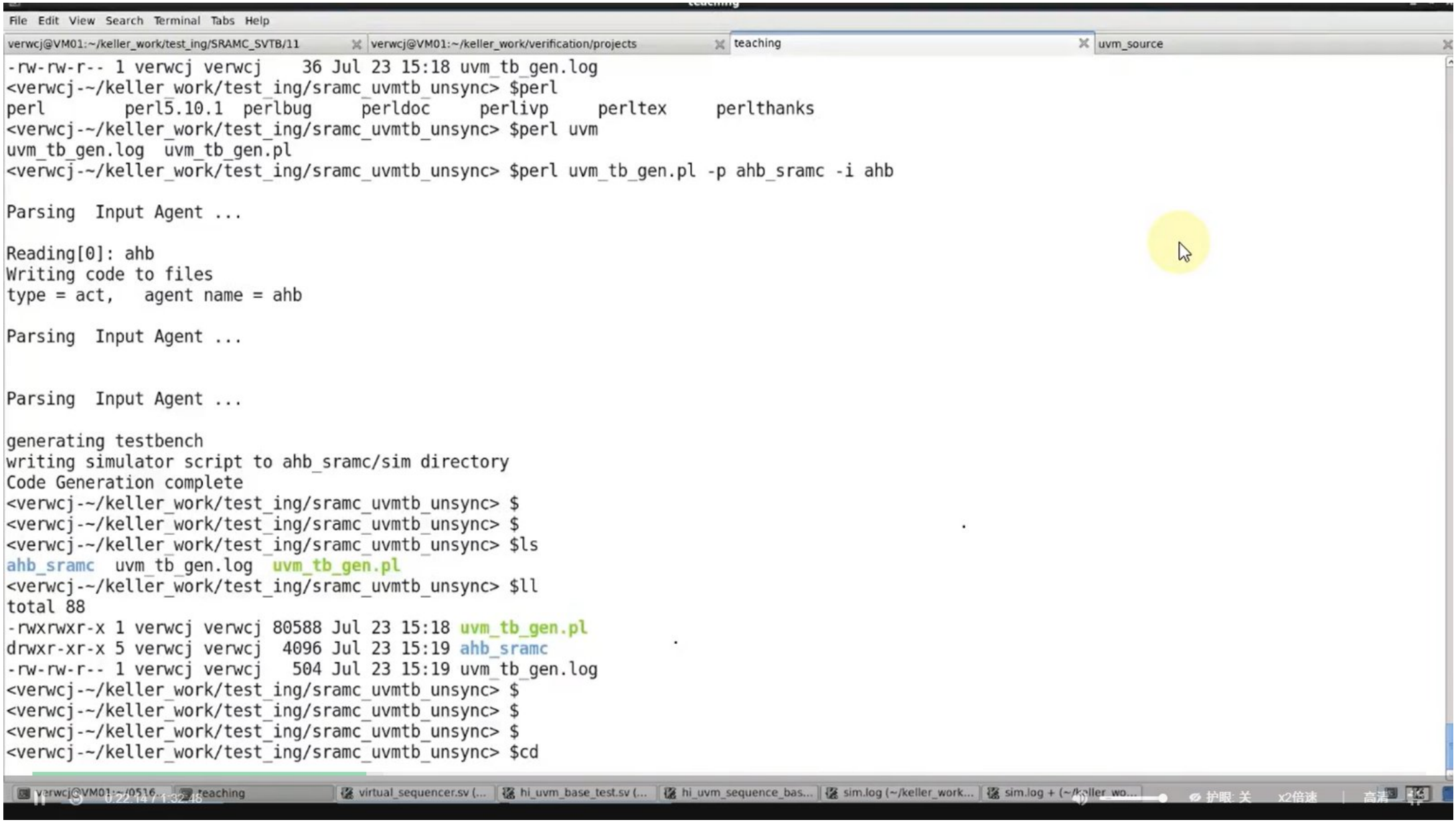
0:12:58 / 1:32:48 护眼: 关 x2倍速 高清

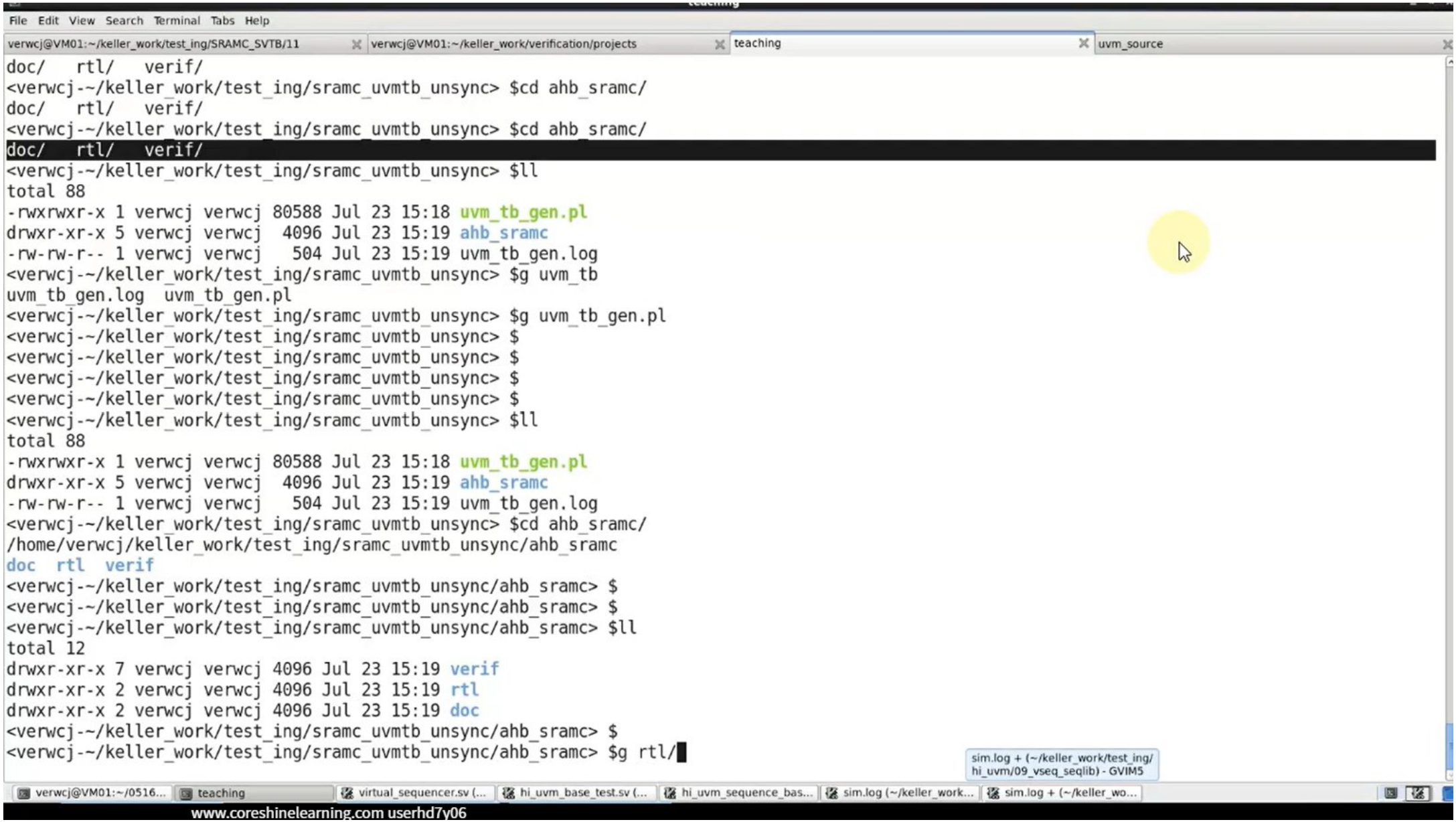

File Edit View Search Terminal Tabs Help
verwcj@VM01:~/keller_work/test_ing/SRAMC_SVTB/11
verwcj@VM01:~/keller_work/verification/projects
teaching
uvm_source
<verwcj-~/keller_work/test_ing> $ll
total 48
drwxrwxr-x 6 verwcj verwcj 4096 Jun 4 08:57 my_test
drwxrwxr-x 3 verwcj verwcj 4096 Jun 4 08:57 seq_check
drwxrwxr-x 12 verwcj verwcj 4096 Jun 4 08:57 router
drwxrwxr-x 5 verwcj verwcj 4096 Jun 4 08:57 demo
-rw-rw-r-- 1 verwcj verwcj 222 Jun 4 08:57 '
drwxrwxr-x 5 verwcj verwcj 4096 Jun 4 08:57 gray_cnt
drwxrwxr-x 4 verwcj verwcj 4096 Jun 4 08:57 full_add
drwxrwxr-x 4 verwcj verwcj 4096 Jun 4 08:57 AFIFO
drwxrwxr-x 5 verwcj verwcj 4096 Jun 4 08:57 oop
drwxrwxr-x 15 verwcj verwcj 4096 Jun 18 16:01 SRAMC_SVTB
drwxrwxr-x 11 verwcj verwcj 4096 Jul 16 15:59 hi_uvm
drwxrwxr-x 2 verwcj verwcj 4096 Jul 23 15:13 sramc_uvmtb_unsync
<verwcj-~/keller_work/test_ing> $cd sramc_uvmtb_unsync
/home/verwcj/keller_work/test_ing/sramc_uvmtb_unsync
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $g /home/verwcj/keller_work/verification/projects/
ahb_flashc/ ahb_sramc_uvmtb/ scripts/ uvm_apb_spi/ uvm_sdmem/ uvm_tb_gen.pl
ahb_sramc/ ahb_sramc_uvmtb.tar sdhost/ uvm_eflash/ uvm_spim/ uvm_tb_gen_sync.pl
ahb_sramc_2203/ ahb_sram_svtb/ sramc_1/ uvm_eflash_regression/ uvm_sramc/
ahb_sramc_2203_sync/ docs/ svtb_ahb_sram/ uvm_hdmi_cec/ uvm_sramc_regresssion/
ahb_sramc_2207/ projects_new/ uvm_ahb_nfc/ uvm_sdhost/ uvm_tb_gen.log
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $cd /home/verwcj/keller_work/verification/projects/
/home/verwcj/keller_work/verification/projects
ahb_flashc ahb_sramc_2207 docs sramc_1 uvm_eflash uvm_sdmem uvm_tb_gen.log
ahb_sramc ahb_sramc_uvmtb projects_new svtb_ahb_sram uvm_eflash_regression uvm_spim uvm_tb_gen.pl
ahb_sramc_2203 ahb_sramc_uvmtb.tar scripts uvm_ahb_nfc uvm_hdmi_cec uvm_sramc uvm_tb_gen_sync.pl
ahb_sramc_2203_sync ahb_sram_svtb sdhost uvm_apb_spi uvm_sdhost uvm_sramc_regresssion
<verwcj-~/keller_work/verification/projects> $./uvm_tb_gen.pl
verwcj@VM01:~/0516...
teaching
virtual_sequencer.sv (...
hi_uvm_base_test.sv (...
hi_uvm_sequence_bas...
sim.log (~/keller_work...
sim.log + (~/keller_wo...


File Edit View Search Terminal Tabs Help
verwcj@VM01:~/keller_work/test_ing/SRAMC_SVTB/11
verwcj@VM01:~/keller_work/verification/projects
teaching
uvm_source
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $g /home/verwcj/keller_work/verification/projects/
ahb_flashc/ ahb_sramc_uvmtb/ scripts/ uvm_apb_spi/ uvm_sdmem/ uvm_tb_gen.pl
ahb_sramc/ ahb_sramc_uvmtb.tar sdhost/ uvm_eflash/ uvm_spim/ uvm_tb_gen_sync.pl
ahb_sramc_2203/ ahb_sram_svtb/ sramc_1/ uvm_eflash_regression/ uvm_sramc/
ahb_sramc_2203_sync/ docs/ svtb_ahb_sram/ uvm_hdmi_cec/ uvm_sramc_regresssion/
ahb_sramc_2207/ projects_new/ uvm_ahb_nfc/ uvm_sdhost/ uvm_tb_gen.log
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $cd /home/verwcj/keller_work/verification/projects/
/home/verwcj/keller_work/verification/projects
ahb_flashc ahb_sramc_2207 docs sramc_1 uvm_eflash uvm_sdmem uvm_tb_gen.log
ahb_sramc ahb_sramc_uvmtb projects_new svtb_ahb_sram uvm_eflash_regression uvm_spim uvm_tb_gen.pl
ahb_sramc_2203 ahb_sramc_uvmtb.tar scripts uvm_ahb_nfc uvm_hdmi_cec uvm_sramc uvm_tb_gen_sync.pl
ahb_sramc_2203_sync ahb_sram_svtb sdhost uvm_apb_spi uvm_sdhost uvm_sramc_regresssion
<verwcj-~/keller_work/verification/projects> $./uvm_tb_gen.pl -h
********USAGE: perl uvm_tb_gen.pl -help/h (print this message)********
this scritps support several input agent, output agent, agent with ral
for the ral agent, support adapter for i2c/spi/apb/ahb,not support yet.
Any suggestion, please contact getwkg@163.com
perl uvm_tb_gen.pl -p project name -i list (input agent name)/active -o list (output agent name)/passive -r register agent with ral
example 1: one input agent, one output agent,one ral agent:
perl uvm_tb_gen.pl -p uart -i uart -o localbus -r apb
example 2: two input agent:
perl uvm_tb_gen.pl -p uart -i uart localbus
example 3: two input agent, one output agent,two ral agent:
perl uvm_tb_gen.pl -p uart -i uart localbus -o outp -r apb ahb
****
<verwcj-~/keller_work/verification/projects> $
verwcj@VM01:~/0516...
teaching
virtual_sequencer.sv (...
hi_uvm_base_test.sv (...
hi_uvm_sequence_bas...
sim.log (~/keller_work...
sim.log + (~/keller_wo...

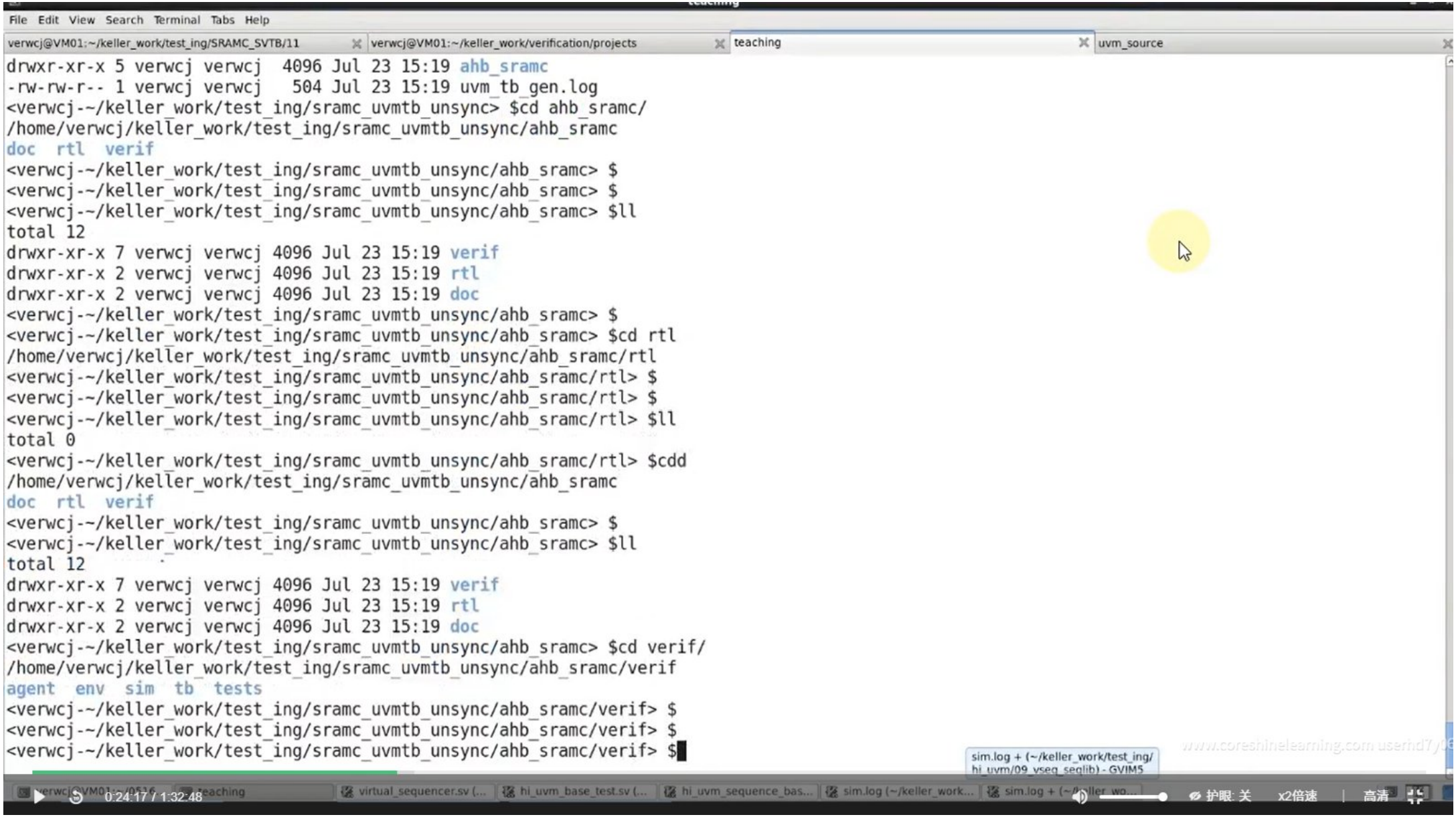

File Edit View Search Terminal Tabs Help
verwcj@VM01:~/keller_work/test_ing/SRAMC_SVTB/11
verwcj@VM01:~/keller_work/verification/projects
teaching
uvm_source
perl uvm_tb_gen.pl -p uart -i uart -o localbus -r apb
example 2: two input agent:
perl uvm_tb_gen.pl -p uart -i uart localbus
example 3: two input agent, one output agent,two ral agent:
perl uvm_tb_gen.pl -p uart -i uart localbus -o outp -r apb ahb
<verwcj-~/keller_work/verification/projects> $
<verwcj-~/keller_work/verification/projects> $
<verwcj-~/keller_work/verification/projects> $
<verwcj-~/keller_work/verification/projects> $
<verwcj-~/keller_work/verification/projects> $pwd
/home/verwcj/keller_work/verification/projects
<verwcj-~/keller_work/verification/projects> $
<verwcj-~/keller_work/verification/projects> $
<verwcj-~/keller_work/verification/projects> $cd
/home/verwcj
0516 Desktop keller_230601.tar.gz keller_work
<verwcj-~> $cd keller
keller_230601.tar.gz keller_work/
<verwcj-~> $cd keller_work/test_ing/
demo/ gray_cnt/ my_test/ router/ SRAMC_SVTB/
AFIFO/ full_add/ hi_uvm/ oop/ seq_check/ sramc_uvmtb_unsync/
<verwcj-~> $cd keller_work/test_ing/sramc
SRAMC_SVTB/ sramc_uvmtb_unsync/
<verwcj-~> $cd keller_work/test_ing/sramc_uvmtb_unsync/
/home/verwcj/keller_work/test_ing/sramc_uvmtb_unsync
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $cp -rf /home/verwcj/keller_work/verification/projects/
ahb_flashc/ ahb_sramc_uvmtb/ scripts/ uvm_apb_spi/ uvm_sdmem/ uvm_tb_gen.pl
ahb_sramc/ ahb_sramc_uvmtb.tar sdhost/ uvm_eflash/ uvm_spim/ uvm_tb_gen_sync.pl
ahb_sramc_2203/ ahb_sram_svtb/ sramc_1/ uvm_eflash_regression/ uvm_sramc/
ahb_sramc_2203_sync/ docs/ svtb_ahb_sram/ uvm_hdmi_cec/ uvm_sramc_regresssion/
ahb_sramc_2207/ projects_new/ uvm_ahb_nfc/ uvm_sdhost/ uvm_tb_gen.log
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $cp -rf /home/verwcj/keller_work/verification/projects/uvm_tb_gen.pl
0:20:05 / 1:32:48
护眼: 关
x2倍速
高清

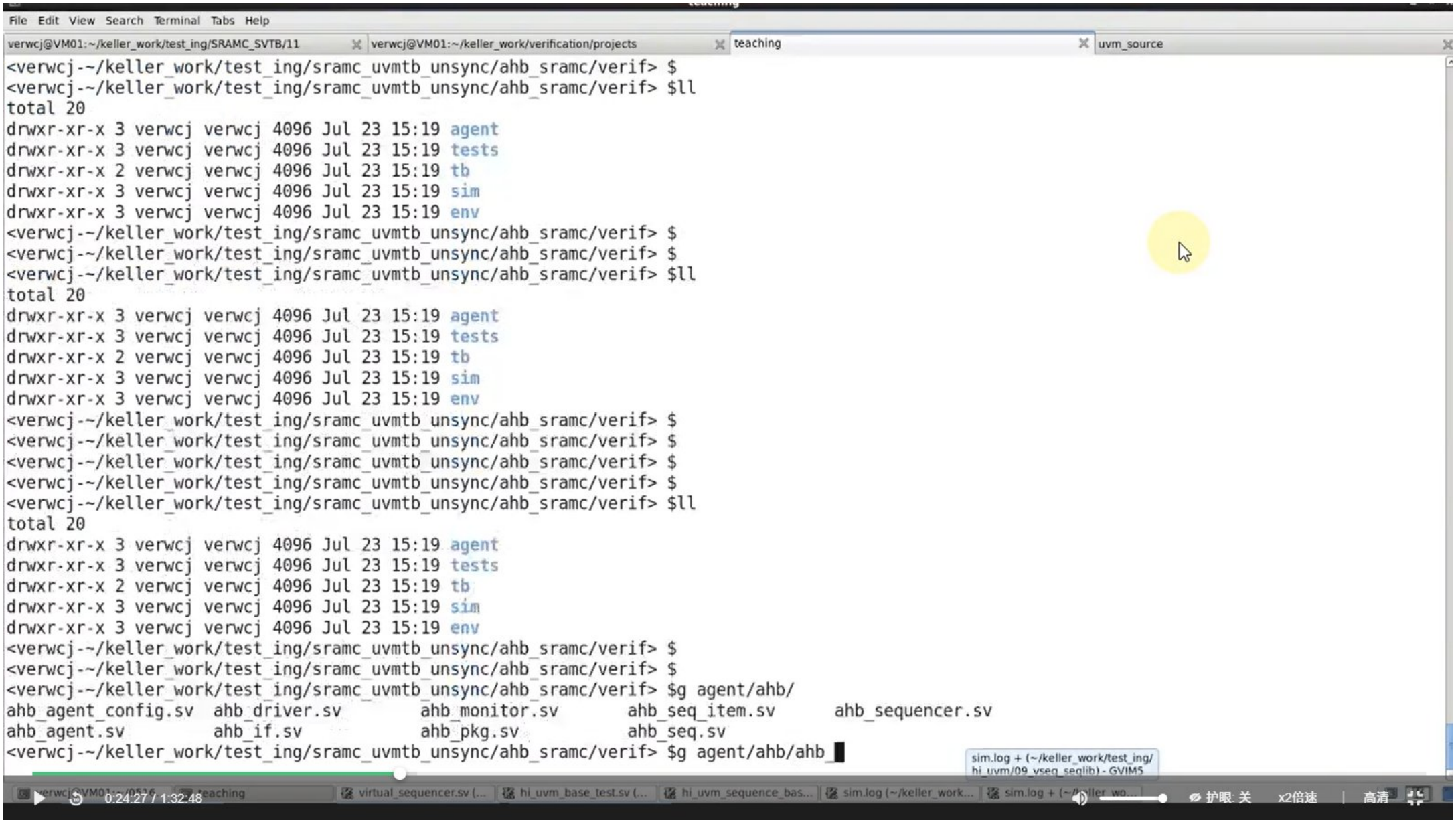

File Edit View Search Terminal Tabs Help
verwcj@VM01:~/keller_work/test_ing/SRAMC_SVTB/11
verwcj@VM01:~/keller_work/verification/projects
teaching
uvm_source
ahb_sramc_2207/ projects_new/ uvm_ahb_nfc/ uvm_sdhost/ uvm_tb_gen.log
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $cp -rf /home/verwcj/keller_work/verification/projects/uvm_tb_gen.pl .
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $ll
total 80
-rwxrwxr-x 1 verwcj verwcj 80588 Jul 23 15:18 uvm_tb_gen.pl
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $./uvm_tb_gen.pl -h
********USAGE: perl uvm_tb_gen.pl -help/h (print this message)********
this scritps support several input agent, output agent, agent with ral
for the ral agent, support adapter for i2c/spi/apb/ahb,not support yet.
Any suggestion, please contact getwkg@163.com
perl uvm_tb_gen.pl -p project name -i list (input agent name)/active -o list (output agent name)/passive -r register agent with ral
example 1: one input agent, one output agent,one ral agent:
perl uvm_tb_gen.pl -p uart -i uart -o localbus -r apb
example 2: two input agent:
perl uvm_tb_gen.pl -p uart -i uart localbus
example 3: two input agent, one output agent,two ral agent:
perl uvm_tb_gen.pl -p uart -i uart localbus -o outp -r apb ahb
****
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
verwcj@VM01:~/0516...
teaching
virtual_sequencer.sv (...
hi_uvm_base_test.sv (...
hi_uvm_sequence_bas...
sim.log (~/keller_work...
sim.log + (~/keller_wo...

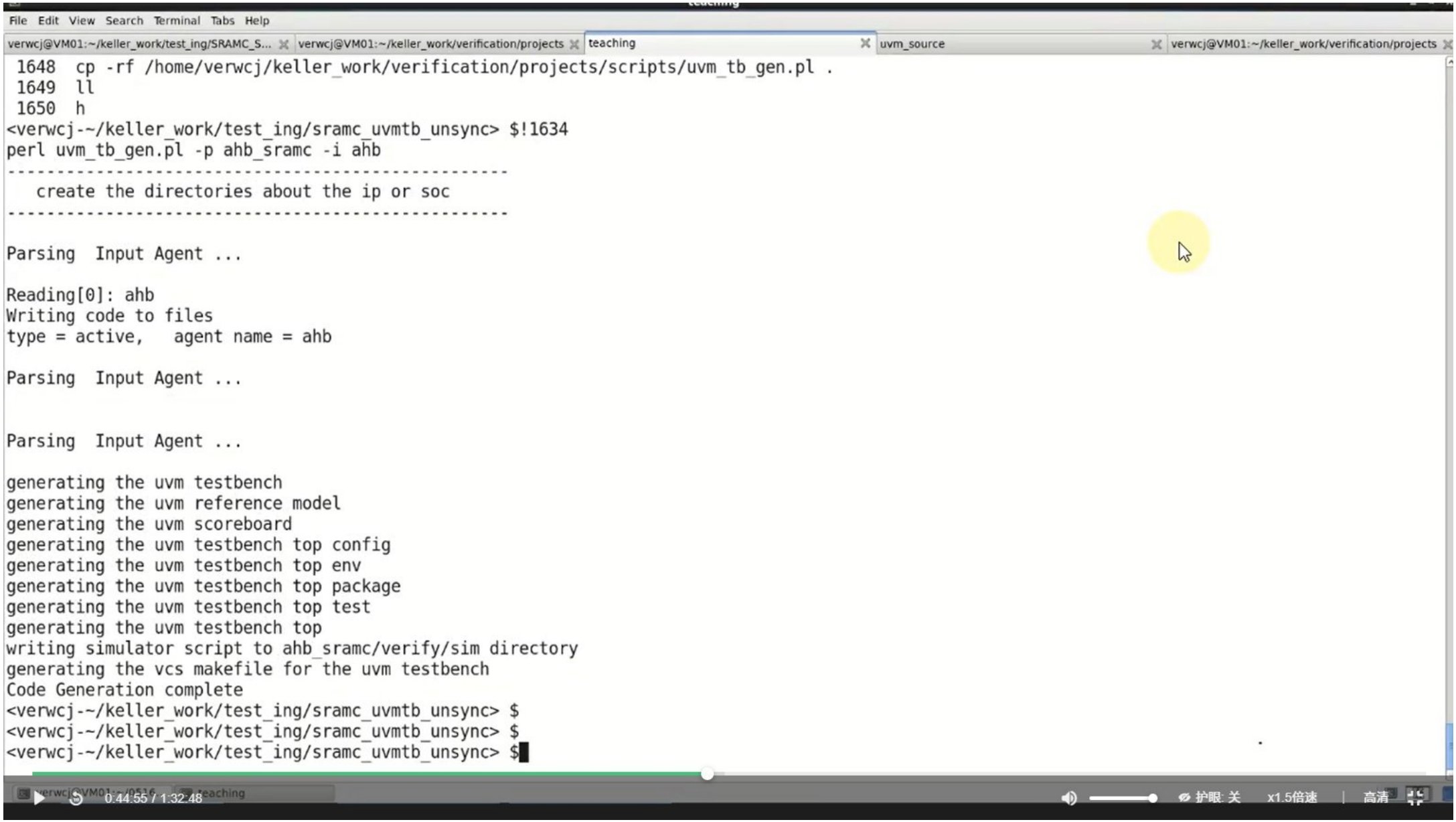

File Edit View Search Terminal Tabs Help
verwcj@VM01:~/keller_work/test_ing/SRAMC_SVTB/11
verwcj@VM01:~/keller_work/verification/projects
teaching
uvm_source
-rw-rw-r-- 1 verwcj verwcj 36 Jul 23 15:18 uvm_tb_gen.log
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $perl
perl perl5.10.1 perlbug perldoc perlivp perltex perlthanks
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $perl uvm
uvm_tb_gen.log uvm_tb_gen.pl
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $perl uvm_tb_gen.pl -p ahb_sramc -i ahb
Parsing Input Agent ...
Reading[0]: ahb
Writing code to files
type = act, agent name = ahb
Parsing Input Agent ...
Parsing Input Agent ...
generating testbench
writing simulator script to ahb_sramc/sim directory
Code Generation complete
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $ls
ahb_sramc uvm_tb_gen.log uvm_tb_gen.pl
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $ll
total 88
-rwxrwxr-x 1 verwcj verwcj 80588 Jul 23 15:18 uvm_tb_gen.pl
drwxr-xr-x 5 verwcj verwcj 4096 Jul 23 15:19 ahb_sramc
-rw-rw-r-- 1 verwcj verwcj 504 Jul 23 15:19 uvm_tb_gen.log
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $cd
teaching
virtual_sequencer.sv (...
hi_uvm_base_test.sv (...
hi_uvm_sequence_bas...
sim.log (~/keller_work...
sim.log + (~/keller_wo...
护眼 关
x2倍速
高清

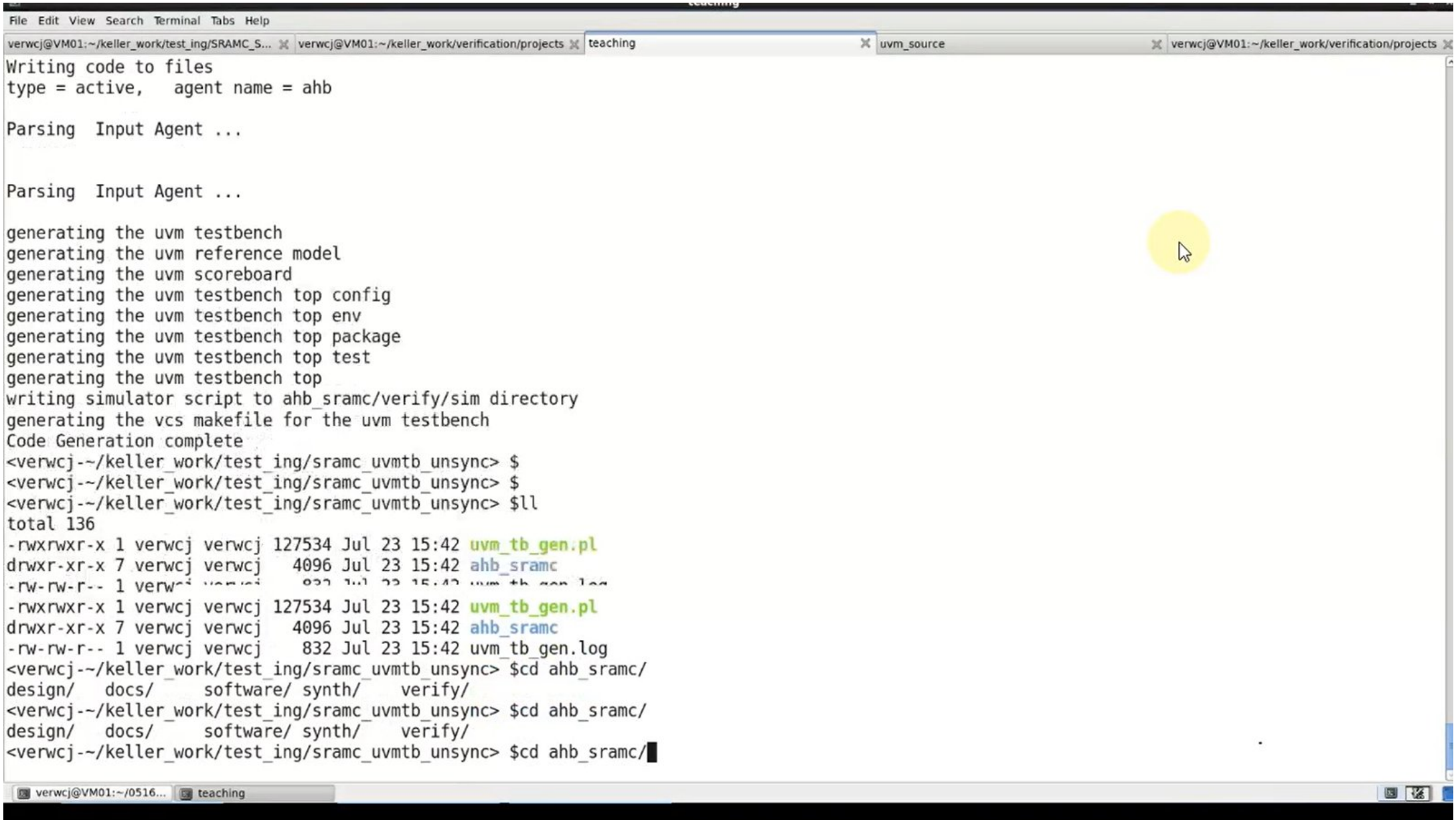


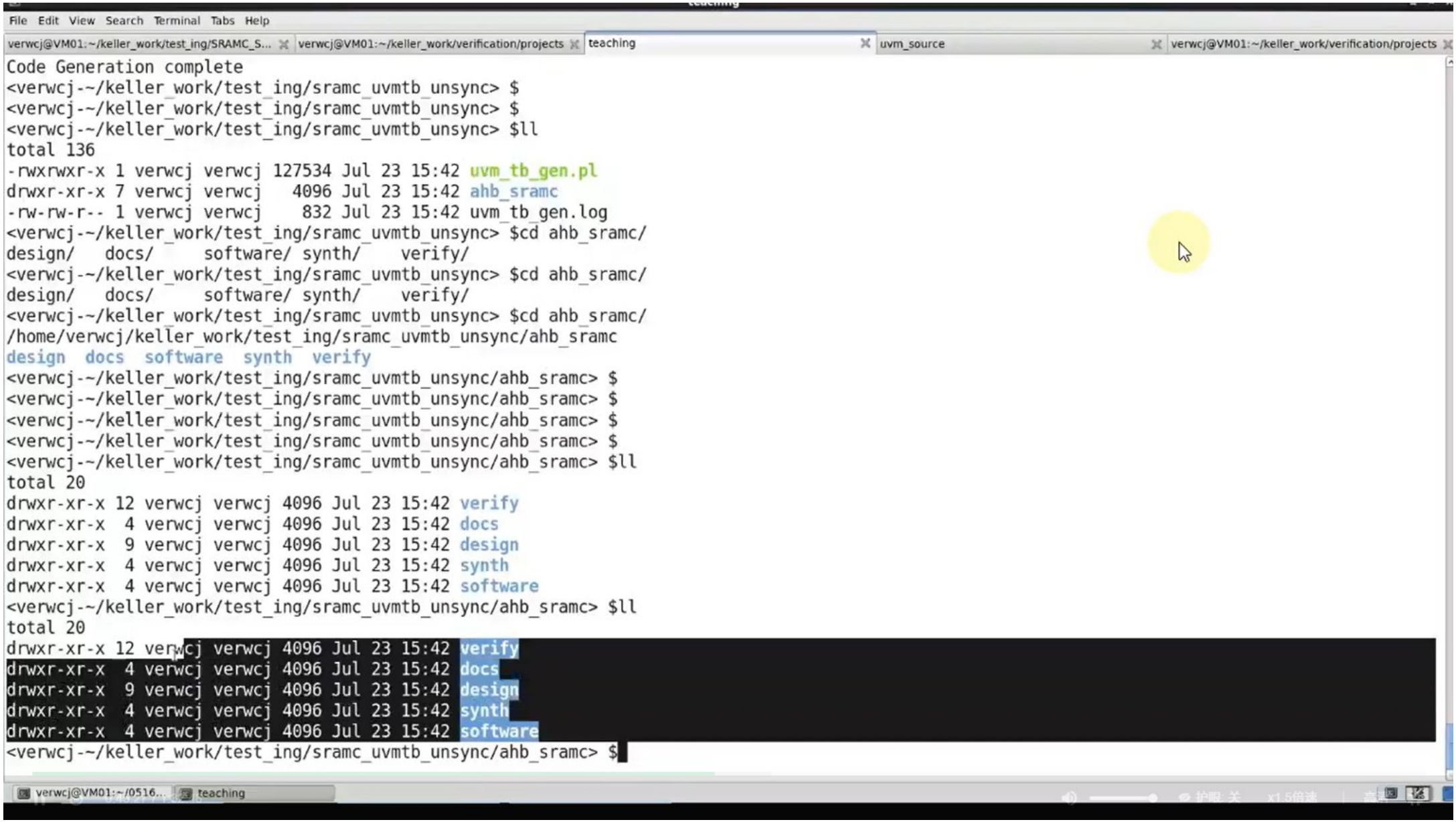
www.coreshinelearning.com userhd7y06

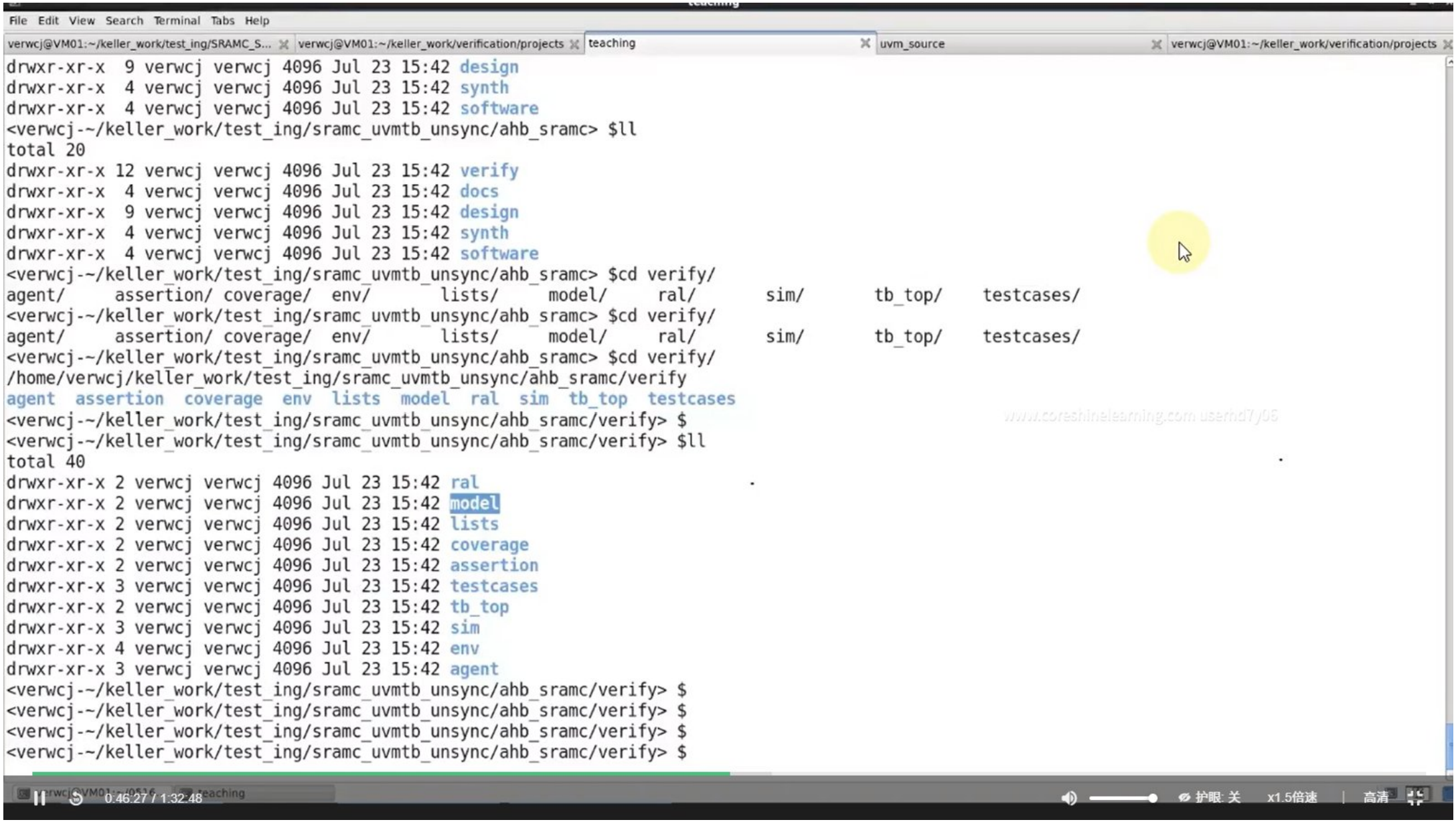

File Edit View Search Terminal Tabs Help
verwcj@VM01:~/keller_work/test_ing/SRAMC_SVTB/11
verwcj@VM01:~/keller_work/verification/projects
teaching
uvm_source
drwxr-xr-x 5 verwcj verwcj 4096 Jul 23 15:19 ahb_sramc
-rw-rw-r-- 1 verwcj verwcj 504 Jul 23 15:19 uvm_tb_gen.log
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $cd ahb_sramc/
/home/verwcj/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc
doc rtl verif
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $ll
total 12
drwxr-xr-x 7 verwcj verwcj 4096 Jul 23 15:19 verif
drwxr-xr-x 2 verwcj verwcj 4096 Jul 23 15:19 rtl
drwxr-xr-x 2 verwcj verwcj 4096 Jul 23 15:19 doc
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $cd rtl
/home/verwcj/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/rtl
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/rtl> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/rtl> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/rtl> $ll
total 0
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/rtl> $cdd
/home/verwcj/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc
doc rtl verif
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $ll
total 12
drwxr-xr-x 7 verwcj verwcj 4096 Jul 23 15:19 verif
drwxr-xr-x 2 verwcj verwcj 4096 Jul 23 15:19 rtl
drwxr-xr-x 2 verwcj verwcj 4096 Jul 23 15:19 doc
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $cd verif/
/home/verwcj/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verif
agent env sim tb tests
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verif> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verif> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verif> $
sim.log + (~/keller_work/test_ing/hi_uvm/09_vseq_seqlib) - GVIM5
0:24:17 / 1:32:48
护眼: 关
x2倍速
高清

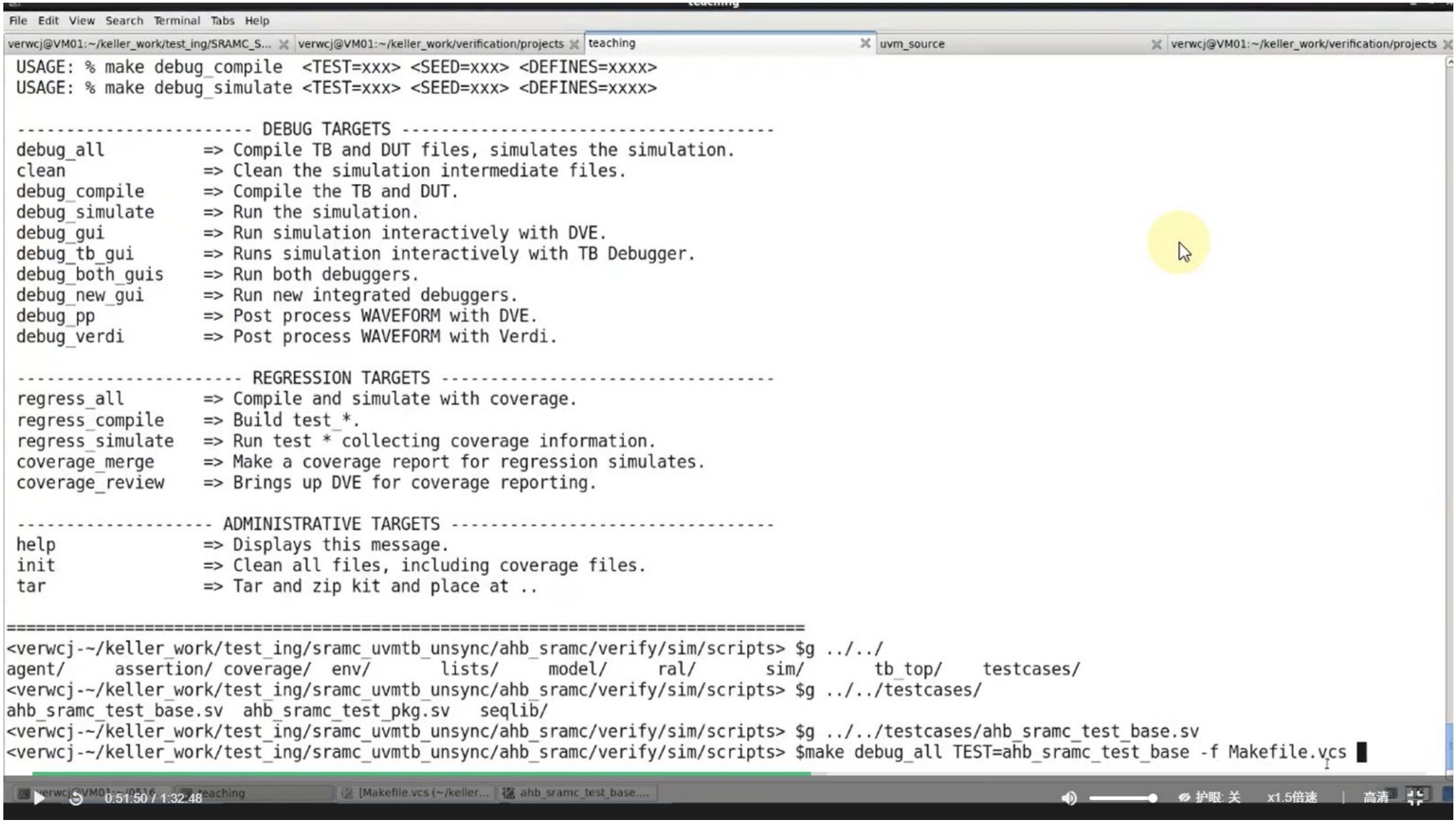

File Edit View Search Terminal Tabs Help
verwcj@VM01:~/keller_work/test_ing/SRAMC_SVTB/11
verwcj@VM01:~/keller_work/verification/projects
teaching
uvm_source
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verif> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verif> $ll
total 20
drwxr-xr-x 3 verwcj verwcj 4096 Jul 23 15:19 agent
drwxr-xr-x 3 verwcj verwcj 4096 Jul 23 15:19 tests
drwxr-xr-x 2 verwcj verwcj 4096 Jul 23 15:19 tb
drwxr-xr-x 3 verwcj verwcj 4096 Jul 23 15:19 sim
drwxr-xr-x 3 verwcj verwcj 4096 Jul 23 15:19 env
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verif> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verif> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verif> $ll
total 20
drwxr-xr-x 3 verwcj verwcj 4096 Jul 23 15:19 agent
drwxr-xr-x 3 verwcj verwcj 4096 Jul 23 15:19 tests
drwxr-xr-x 2 verwcj verwcj 4096 Jul 23 15:19 tb
drwxr-xr-x 3 verwcj verwcj 4096 Jul 23 15:19 sim
drwxr-xr-x 3 verwcj verwcj 4096 Jul 23 15:19 env
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verif> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verif> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verif> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verif> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verif> $ll
total 20
drwxr-xr-x 3 verwcj verwcj 4096 Jul 23 15:19 agent
drwxr-xr-x 3 verwcj verwcj 4096 Jul 23 15:19 tests
drwxr-xr-x 2 verwcj verwcj 4096 Jul 23 15:19 tb
drwxr-xr-x 3 verwcj verwcj 4096 Jul 23 15:19 sim
drwxr-xr-x 3 verwcj verwcj 4096 Jul 23 15:19 env
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verif> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verif> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verif> $g agent/ahb/
ahb_agent_config.sv ahb_driver.sv ahb_monitor.sv ahb_seq_item.sv ahb_sequencer.sv
ahb_agent.sv ahb_if.sv ahb_pkg.sv ahb_seq.sv
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verif> $g agent/ahb/ahb_
sim.log + (~/keller_work/test_ing/hi_uvm/09_vseq_seqlib) - GVIM5
0:24:27 / 1:32:48
护眼: 关
x2倍速
高清

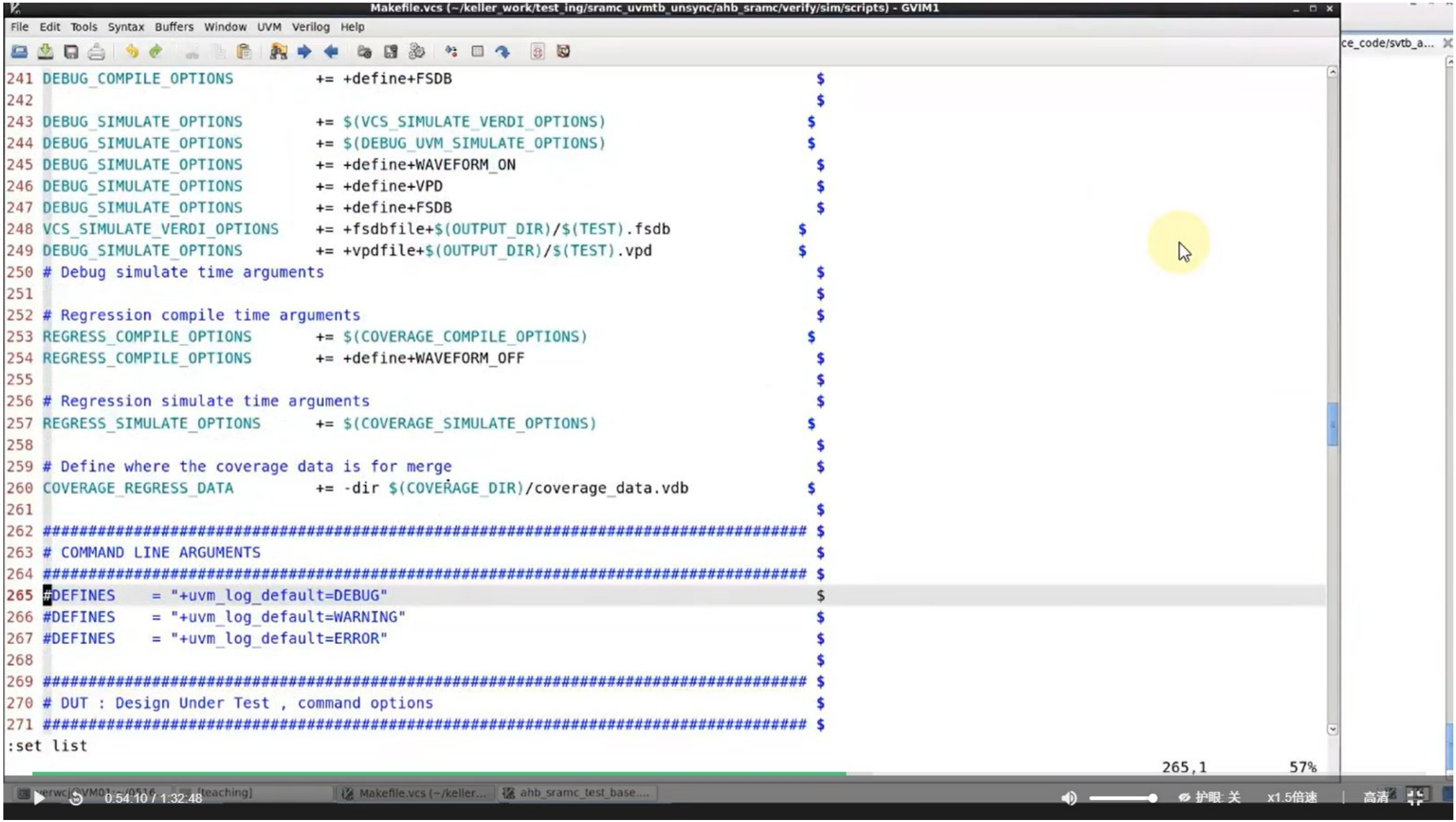

File Edit View Search Terminal Tabs Help
verwcj@VM01:~/keller_work/test_ing/SRAMC_S...
verwcj@VM01:~/keller_work/verification/projects
teaching
uvm_source
verwcj@VM01:~/keller_work/verification/projects
1648 cp -rf /home/verwcj/keller_work/verification/projects/scripts/uvm_tb_gen.pl .
1649 ll
1650 h
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $!1634
perl uvm_tb_gen.pl -p ahb_sramc -i ahb
create the directories about the ip or soc
Parsing Input Agent ...
Reading[0]: ahb
Writing code to files
type = active, agent name = ahb
Parsing Input Agent ...
Parsing Input Agent ...
generating the uvm testbench
generating the uvm reference model
generating the uvm scoreboard
generating the uvm testbench top config
generating the uvm testbench top env
generating the uvm testbench top package
generating the uvm testbench top test
generating the uvm testbench top
writing simulator script to ahb_sramc/verify/sim directory
generating the vcs makefile for the uvm testbench
Code Generation complete
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
0:44:55 / 1:32:48
护眼: 关
x1.5倍速
高清

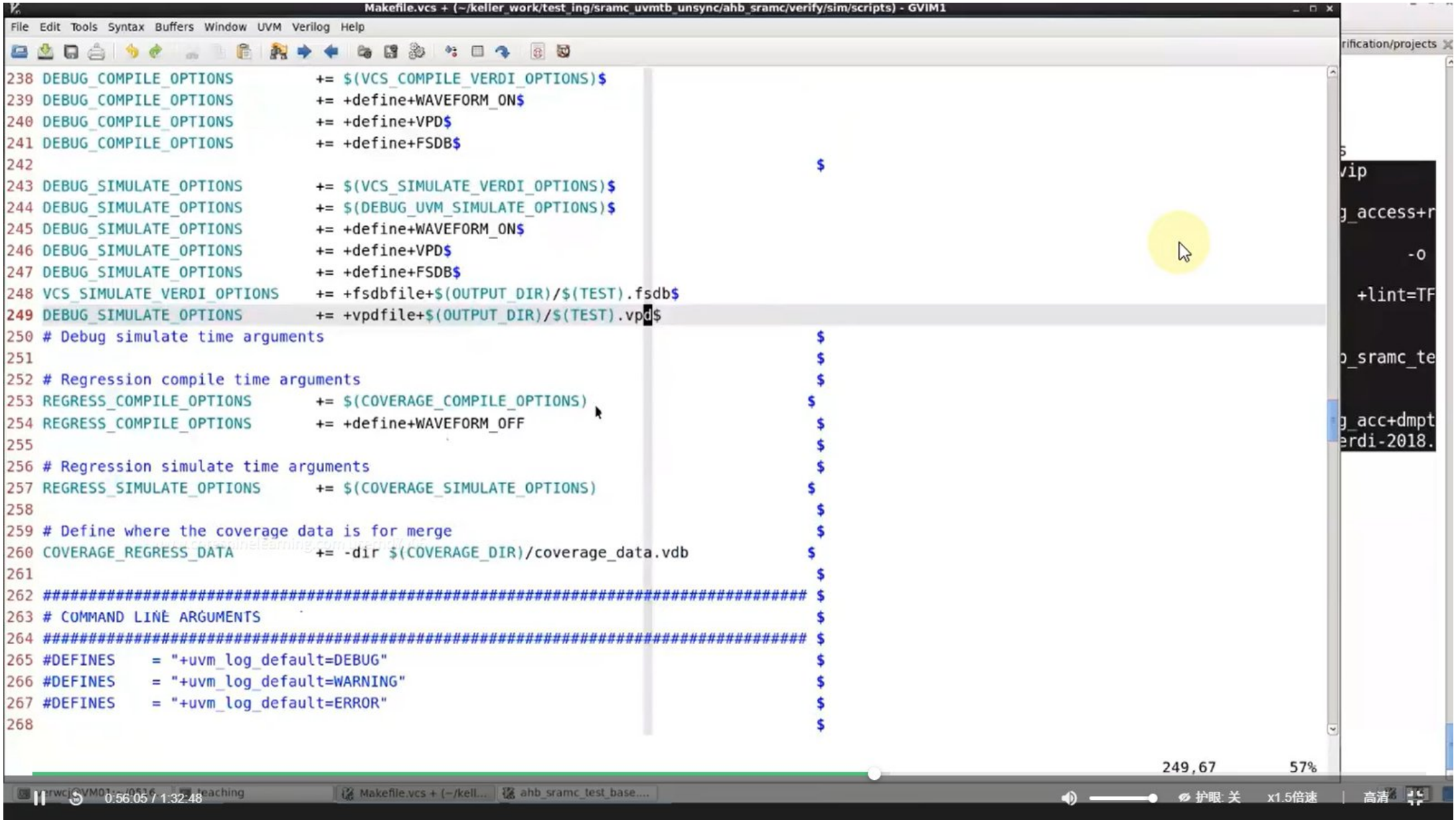

File Edit View Search Terminal Tabs Help
verwcj@VM01:~/keller_work/test_ing/SRAMC_S...
verwcj@VM01:~/keller_work/verification/projects
teaching
uvm_source
verwcj@VM01:~/keller_work/verification/projects
Writing code to files
type = active, agent name = ahb
Parsing Input Agent ...
Parsing Input Agent ...
generating the uvm testbench
generating the uvm reference model
generating the uvm scoreboard
generating the uvm testbench top config
generating the uvm testbench top env
generating the uvm testbench top package
generating the uvm testbench top test
generating the uvm testbench top
writing simulator script to ahb_sramc/verify/sim directory
generating the vcs makefile for the uvm testbench
Code Generation complete
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $ll
total 136
-rwxrwxr-x 1 verwcj verwcj 127534 Jul 23 15:42 uvm_tb_gen.pl
drwxr-xr-x 7 verwcj verwcj   4096 Jul 23 15:42 ahb_sramc
-rwxrwxr-x 1 verwcj verwcj 127534 Jul 23 15:42 uvm_tb_gen.pl
drwxr-xr-x 7 verwcj verwcj   4096 Jul 23 15:42 ahb_sramc
-rw-rw-r-- 1 verwcj verwcj   832 Jul 23 15:42 uvm_tb_gen.log
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $cd ahb_sramc/
design/ docs/ software/ synth/ verify/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $cd ahb_sramc/
design/ docs/ software/ synth/ verify/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $cd ahb_sramc/
verwcj@VM01:~/0516...
teaching

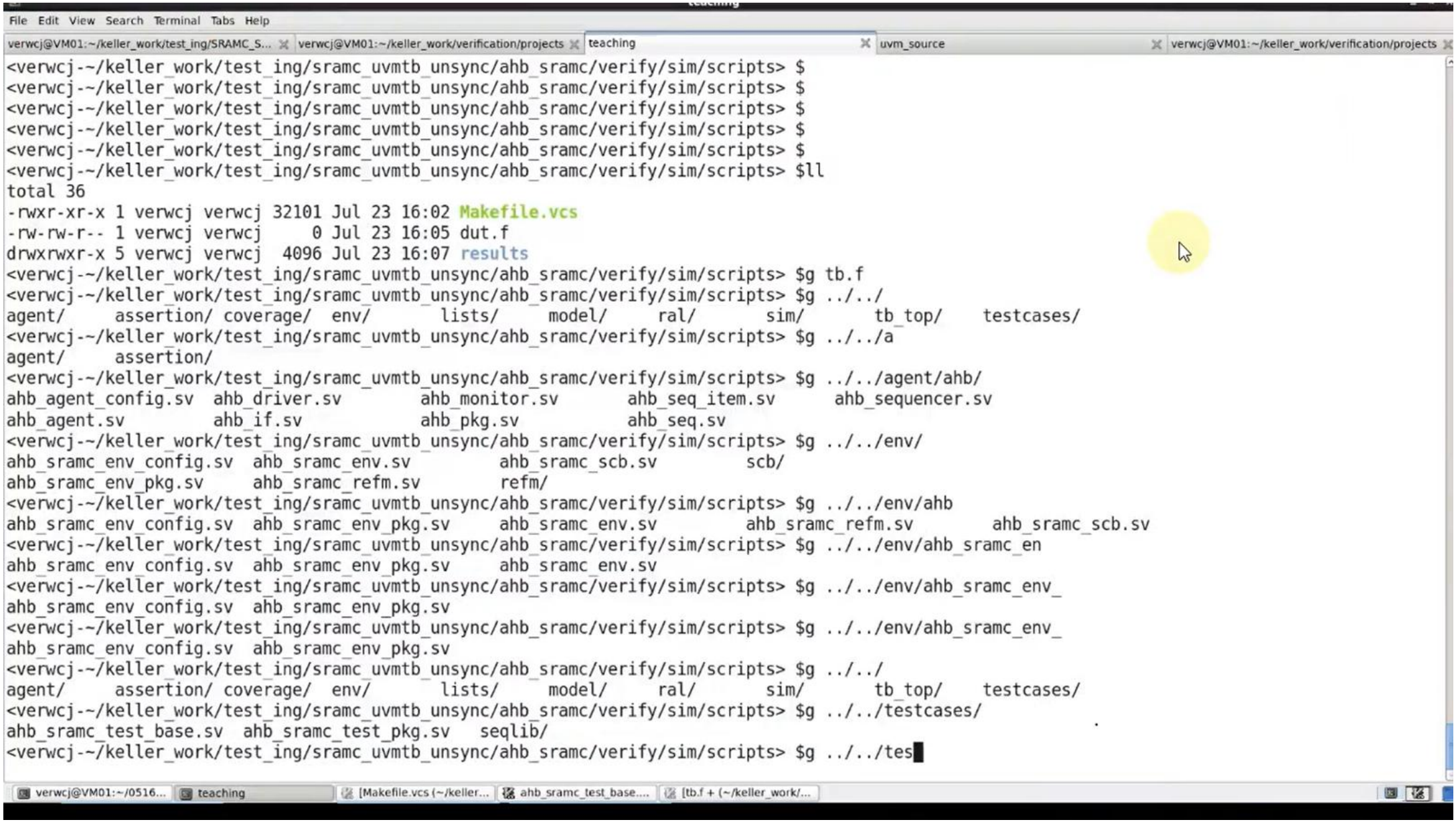

File Edit View Search Terminal Tabs Help
verwcj@VM01:~/keller_work/test_ing/SRAMC_S...
verwcj@VM01:~/keller_work/verification/projects
teaching
uvm_source
verwcj@VM01:~/keller_work/verification/projects
Code Generation complete
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $ll
total 136
-rwxrwxr-x 1 verwcj verwcj 127534 Jul 23 15:42 uvm_tb_gen.pl
drwxr-xr-x 7 verwcj verwcj 4096 Jul 23 15:42 ahb_sramc
-rw-rw-r-- 1 verwcj verwcj 832 Jul 23 15:42 uvm_tb_gen.log
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $cd ahb_sramc/
design/ docs/ software/ synth/ verify/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $cd ahb_sramc/
design/ docs/ software/ synth/ verify/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync> $cd ahb_sramc/
/home/verwcj/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc
design docs software synth verify
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $ll
total 20
drwxr-xr-x 12 verwcj verwcj 4096 Jul 23 15:42 verify
drwxr-xr-x 4 verwcj verwcj 4096 Jul 23 15:42 docs
drwxr-xr-x 9 verwcj verwcj 4096 Jul 23 15:42 design
drwxr-xr-x 4 verwcj verwcj 4096 Jul 23 15:42 synth
drwxr-xr-x 4 verwcj verwcj 4096 Jul 23 15:42 software
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $ll
total 20
drwxr-xr-x 12 verwcj verwcj 4096 Jul 23 15:42 verify
drwxr-xr-x 4 verwcj verwcj 4096 Jul 23 15:42 docs
drwxr-xr-x 9 verwcj verwcj 4096 Jul 23 15:42 design
drwxr-xr-x 4 verwcj verwcj 4096 Jul 23 15:42 synth
drwxr-xr-x 4 verwcj verwcj 4096 Jul 23 15:42 software
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $
verwcj@VM01:~/0516...
teaching

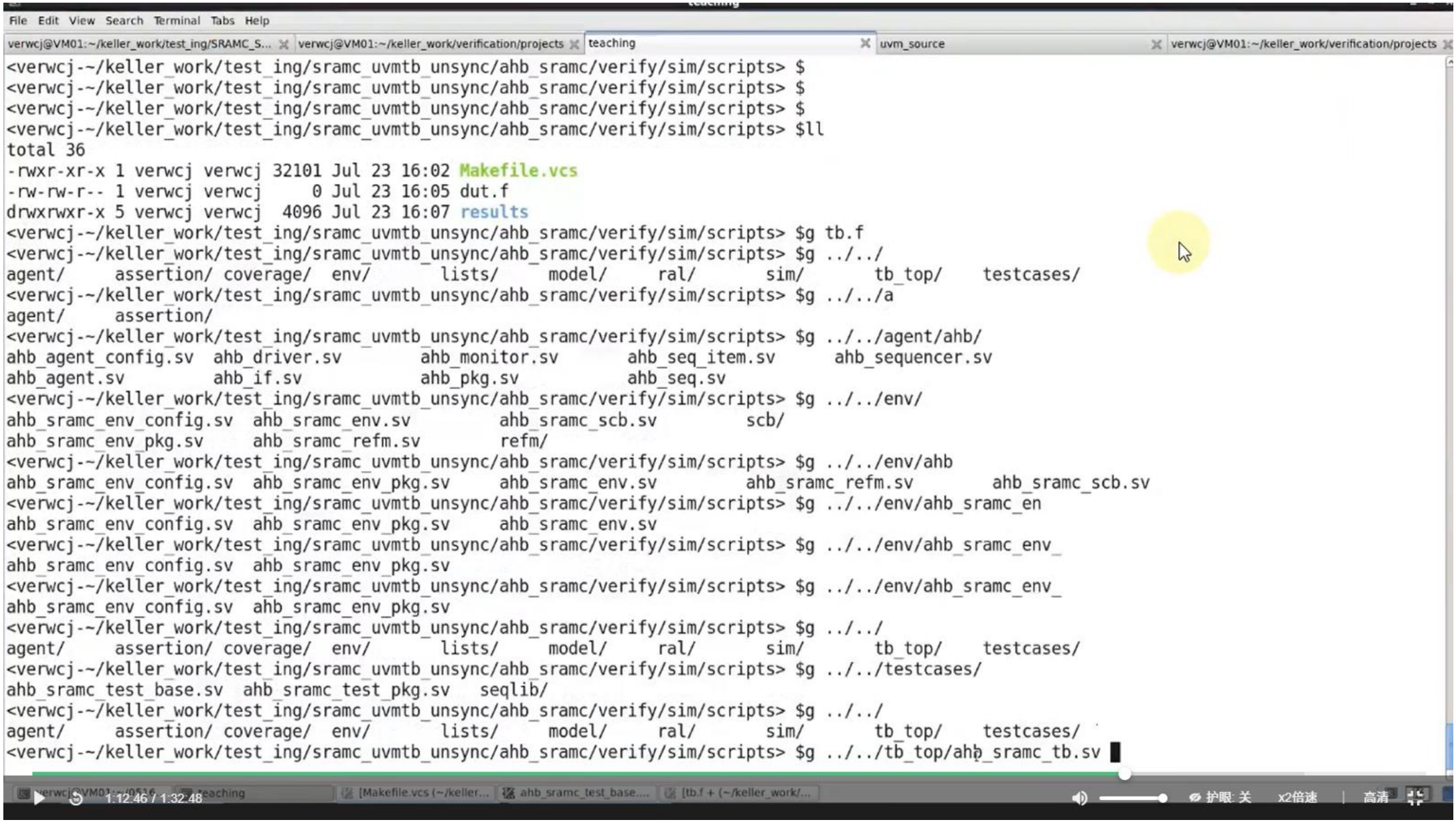

File Edit View Search Terminal Tabs Help
verwcj@VM01:~/keller_work/test_ing/SRAMC_S...
verwcj@VM01:~/keller_work/verification/projects
teaching
uvm_source
verwcj@VM01:~/keller_work/verification/projects
drwxr-xr-x 9 verwcj verwcj 4096 Jul 23 15:42 design
drwxr-xr-x 4 verwcj verwcj 4096 Jul 23 15:42 synth
drwxr-xr-x 4 verwcj verwcj 4096 Jul 23 15:42 software
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $ll
total 20
drwxr-xr-x 12 verwcj verwcj 4096 Jul 23 15:42 verify
drwxr-xr-x 4 verwcj verwcj 4096 Jul 23 15:42 docs
drwxr-xr-x 9 verwcj verwcj 4096 Jul 23 15:42 design
drwxr-xr-x 4 verwcj verwcj 4096 Jul 23 15:42 synth
drwxr-xr-x 4 verwcj verwcj 4096 Jul 23 15:42 software
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $cd verify/
agent/ assertion/ coverage/ env/ lists/ model/ ral/ sim/ tb_top/ testcases/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $cd verify/
agent/ assertion/ coverage/ env/ lists/ model/ ral/ sim/ tb_top/ testcases/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc> $cd verify/
/home/verwcj/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify
agent assertion coverage env lists model ral sim tb_top testcases
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify> $ll
total 40
drwxr-xr-x 2 verwcj verwcj 4096 Jul 23 15:42 ral
drwxr-xr-x 2 verwcj verwcj 4096 Jul 23 15:42 model
drwxr-xr-x 2 verwcj verwcj 4096 Jul 23 15:42 lists
drwxr-xr-x 2 verwcj verwcj 4096 Jul 23 15:42 coverage
drwxr-xr-x 2 verwcj verwcj 4096 Jul 23 15:42 assertion
drwxr-xr-x 3 verwcj verwcj 4096 Jul 23 15:42 testcases
drwxr-xr-x 2 verwcj verwcj 4096 Jul 23 15:42 tb_top
drwxr-xr-x 3 verwcj verwcj 4096 Jul 23 15:42 sim
drwxr-xr-x 4 verwcj verwcj 4096 Jul 23 15:42 env
drwxr-xr-x 3 verwcj verwcj 4096 Jul 23 15:42 agent
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify> $
0:46:27 / 1:32:48
护眼: 关
x1.5倍速
高清

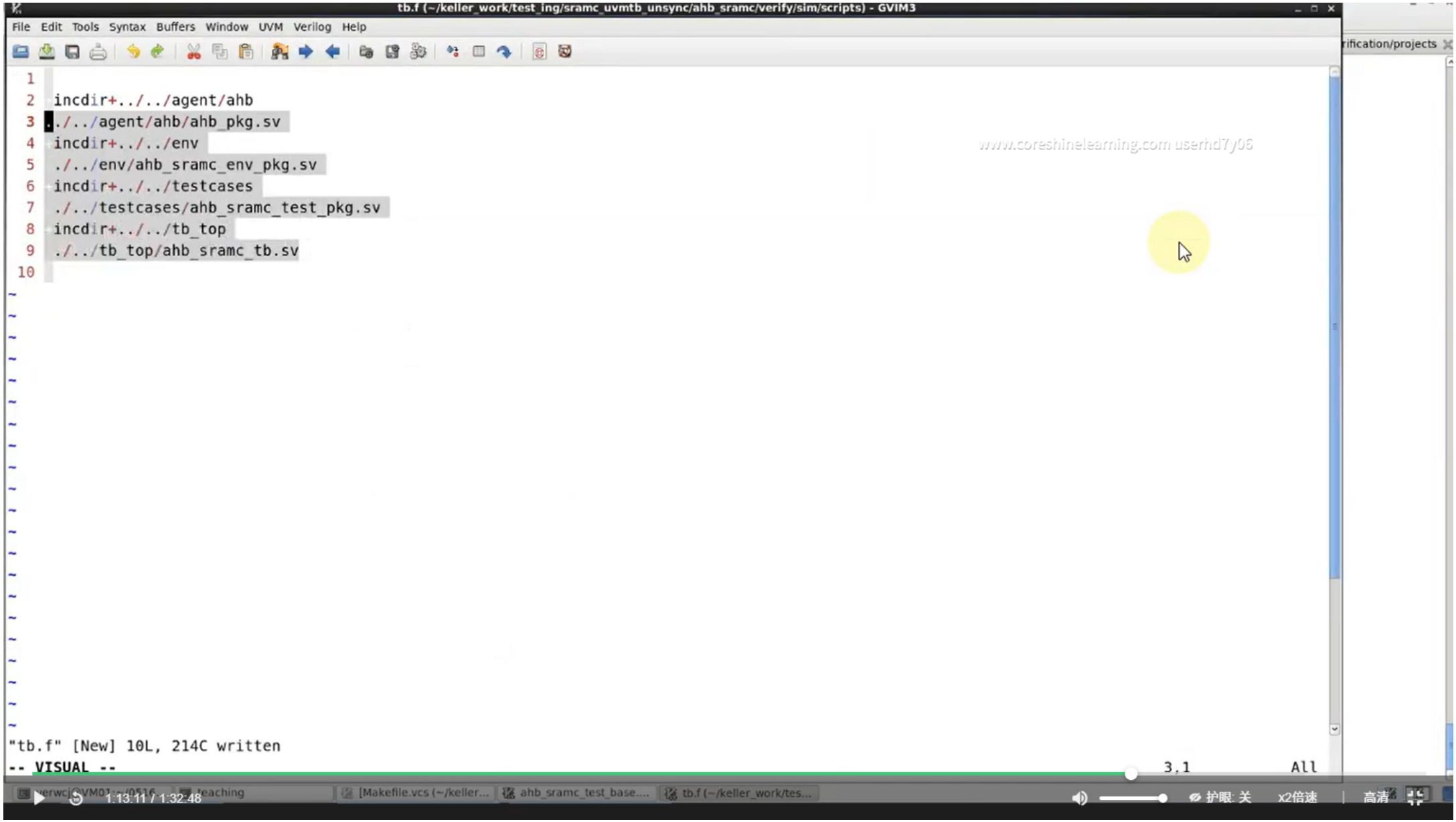

File Edit View Search Terminal Tabs Help
verwcj@VM01:~/keller_work/test_ing/SRAMC_S...
verwcj@VM01:~/keller_work/verification/projects
teaching
uvm_source
verwcj@VM01:~/keller_work/verification/projects
USAGE: % make debug_compile <TEST=xxx> <SEED=xxx> <DEFINES=xxxx>
USAGE: % make debug_simulate <TEST=xxx> <SEED=xxx> <DEFINES=xxxx>
---------- DEBUG TARGETS ----------
debug_all => Compile TB and DUT files, simulates the simulation.
clean => Clean the simulation intermediate files.
debug_compile => Compile the TB and DUT.
debug_simulate => Run the simulation.
debug_gui => Run simulation interactively with DVE.
debug_tb_gui => Runs simulation interactively with TB Debugger.
debug_both_guis => Run both debuggers.
debug_new_gui => Run new integrated debuggers.
debug_pp => Post process WAVEFORM with DVE.
debug_verdi => Post process WAVEFORM with Verdi.
---------- REGRESSION TARGETS ----------
regress_all => Compile and simulate with coverage.
regress_compile => Build test_*.
regress_simulate => Run test * collecting coverage information.
coverage_merge => Make a coverage report for regression simulates.
coverage_review => Brings up DVE for coverage reporting.
---------- ADMINISTRATIVE TARGETS ----------
help => Displays this message.
init => Clean all files, including coverage files.
tar => Tar and zip kit and place at ..
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../
agent/ assertion/ coverage/ env/ lists/ model/ ral/ sim/ tb_top/ testcases/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../testcases/
ahb_sramc_test_base.sv ahb_sramc_test_pkg.sv seqlib/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../testcases/ahb_sramc_test_base.sv
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $make debug_all TEST=ahb_sramc_test_base -f Makefile.vcs
0:51:50 / 1:32:48
护眼: 关
x1.5倍速
高清


Makefile.vcs (~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts) - GVIM1
File Edit Tools Syntax Buffers Window UVM Verilog Help
DEBUG_COMPILE_OPTIONS += +define+FSDB
DEBUG_SIMULATE_OPTIONS += $(VCS_SIMULATE_VERDI_OPTIONS)
DEBUG_SIMULATE_OPTIONS += $(DEBUG_UVM_SIMULATE_OPTIONS)
DEBUG_SIMULATE_OPTIONS += +define+WAVEFORM_ON
DEBUG_SIMULATE_OPTIONS += +define+VPD
DEBUG_SIMULATE_OPTIONS += +define+FSDB
VCS_SIMULATE_VERDI_OPTIONS += +fsdbfile+$(OUTPUT_DIR)/$(TEST).fsdb
DEBUG_SIMULATE_OPTIONS += +vpdfile+$(OUTPUT_DIR)/$(TEST).vpd
Debug simulate time arguments
Regression compile time arguments
REGRESS_COMPILE_OPTIONS += $(COVERAGE_COMPILE_OPTIONS)
REGRESS_COMPILE_OPTIONS += +define+WAVEFORM_OFF
Regression simulate time arguments
REGRESS_SIMULATE_OPTIONS += $(COVERAGE_SIMULATE_OPTIONS)
Define where the coverage data is for merge
COVERAGE_REGRESS_DATA += -dir $(COVERAGE_DIR)/coverage_data.vdb
COMMAND LINE ARGUMENTS
#DEFINES = "+uvm_log_default=DEBUG"
#DEFINES = "+uvm_log_default=WARNING"
#DEFINES = "+uvm_log_default=ERROR"
DUT : Design Under Test , command options
:set list
265,1 57%
ce_code/svtb_a...
0:54:10 / 1:32:48
Makefile.vcs (~/keller...
ahb_sramc_test_base....
护眼: 关
x1.5倍速
高清


Makefile.vcs + (~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts) - GVIM1
File Edit Tools Syntax Buffers Window UVM Verilog Help
238 DEBUG_COMPILE_OPTIONS += $(VCS_COMPILE_VERDI_OPTIONS)$
239 DEBUG_COMPILE_OPTIONS += +define+WAVEFORM_ON$
240 DEBUG_COMPILE_OPTIONS += +define+VPD$
241 DEBUG_COMPILE_OPTIONS += +define+FSDB$
242 $
243 DEBUG_SIMULATE_OPTIONS += $(VCS_SIMULATE_VERDI_OPTIONS)$
244 DEBUG_SIMULATE_OPTIONS += $(DEBUG_UVM_SIMULATE_OPTIONS)$
245 DEBUG_SIMULATE_OPTIONS += +define+WAVEFORM_ON$
246 DEBUG_SIMULATE_OPTIONS += +define+VPD$
247 DEBUG_SIMULATE_OPTIONS += +define+FSDB$
248 VCS_SIMULATE_VERDI_OPTIONS += +fsdbfile+$(OUTPUT_DIR)/$(TEST).fsdb$
249 DEBUG_SIMULATE_OPTIONS += +vpdfile+$(OUTPUT_DIR)/$(TEST).vpd$
250 # Debug simulate time arguments $
251 $
252 # Regression compile time arguments $
253 REGRESS_COMPILE_OPTIONS += $(COVERAGE_COMPILE_OPTIONS) $
254 REGRESS_COMPILE_OPTIONS += +define+WAVEFORM_OFF $
255 $
256 # Regression simulate time arguments $
257 REGRESS_SIMULATE_OPTIONS += $(COVERAGE_SIMULATE_OPTIONS) $
258 $
259 # Define where the coverage data is for merge $
260 COVERAGE_REGRESS_DATA += -dir $(COVERAGE_DIR)/coverage_data.vdb $
261 $
262 ## $
263 # COMMAND LINE ARGUMENTS $
264 ## $
265 #DEFINES = "+uvm_log_default=DEBUG" $
266 #DEFINES = "+uvm_log_default=WARNING" $
267 #DEFINES = "+uvm_log_default=ERROR" $
268 $
249,67 57%
rification/projects
vip
g_access+r
-o
+lint=TF
o_sramc_te
g_acc+dmpt
erdi-2018.
0:56:05 / 1:32:48
护眼: 关
x1.5倍速
高清


File Edit View Search Terminal Tabs Help
verwcj@VM01:~/keller_work/test_ing/SRAMC_S...
verwcj@VM01:~/keller_work/verification/projects
teaching
uvm_source
verwcj@VM01:~/keller_work/verification/projects
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $ll
total 36
-rwxr-xr-x 1 verwcj verwcj 32101 Jul 23 16:02 Makefile.vcs
-rw-rw-r-- 1 verwcj verwcj   0 Jul 23 16:05 dut.f
drwxrwxr-x 5 verwcj verwcj 4096 Jul 23 16:07 results
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g tb.f
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../
agent/   assertion/ coverage/ env/   lists/   model/   ral/   sim/   tb_top/   testcases/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../a
agent/   assertion/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../agent/ahb/
ahb_agent_config.sv ahb_driver.sv   ahb_monitor.sv   ahb_seq_item.sv   ahb_sequencer.sv
ahb_agent.sv   ahb_if.sv   ahb_pkg.sv   ahb_seq.sv
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../env/
ahb_sramc_env_config.sv ahb_sramc_env.sv   ahb_sramc_scb.sv   scb/
ahb_sramc_env_pkg.sv   ahb_sramc_refm.sv   refm/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../env/ahb
ahb_sramc_env_config.sv ahb_sramc_env_pkg.sv   ahb_sramc_env.sv   ahb_sramc_refm.sv   ahb_sramc_scb.sv
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../env/ahb_sramc_en
ahb_sramc_env_config.sv ahb_sramc_env_pkg.sv   ahb_sramc_env.sv
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../env/ahb_sramc_env_
ahb_sramc_env_config.sv ahb_sramc_env_pkg.sv
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../env/ahb_sramc_env_
ahb_sramc_env_config.sv ahb_sramc_env_pkg.sv
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../
agent/   assertion/ coverage/ env/   lists/   model/   ral/   sim/   tb_top/   testcases/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../testcases/
ahb_sramc_test_base.sv ahb_sramc_test_pkg.sv   seqlib/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../tes
verwcj@VM01:~/0516...
teaching
[Makefile.vcs (~/keller...
ahb_sramc_test_base....
[tb.f + (~/keller_work/...


File Edit View Search Terminal Tabs Help
verwcj@VM01:~/keller_work/test_ing/SRAMC_S...
verwcj@VM01:~/keller_work/verification/projects
teaching
uvm_source
verwcj@VM01:~/keller_work/verification/projects
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $ll
total 36
-rwxr-xr-x 1 verwcj verwcj 32101 Jul 23 16:02 Makefile.vcs
-rw-rw-r-- 1 verwcj verwcj 0 Jul 23 16:05 dut.f
drwxrwxr-x 5 verwcj verwcj 4096 Jul 23 16:07 results
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g tb.f
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../
agent/ assertion/ coverage/ env/ lists/ model/ ral/ sim/ tb_top/ testcases/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../a
agent/ assertion/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../agent/ahb/
ahb_agent_config.sv ahb_driver.sv ahb_monitor.sv ahb_seq_item.sv ahb_sequencer.sv
ahb_agent.sv ahb_if.sv ahb_pkg.sv ahb_seq.sv
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../env/
ahb_sramc_env_config.sv ahb_sramc_env.sv ahb_sramc_scb.sv scb/
ahb_sramc_env_pkg.sv ahb_sramc_refm.sv refm/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../env/ahb
ahb_sramc_env_config.sv ahb_sramc_env_pkg.sv ahb_sramc_env.sv ahb_sramc_refm.sv ahb_sramc_scb.sv
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../env/ahb_sramc_en
ahb_sramc_env_config.sv ahb_sramc_env_pkg.sv ahb_sramc_env.sv
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../env/ahb_sramc_env_
ahb_sramc_env_config.sv ahb_sramc_env_pkg.sv
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../env/ahb_sramc_env_
ahb_sramc_env_config.sv ahb_sramc_env_pkg.sv
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../
agent/ assertion/ coverage/ env/ lists/ model/ ral/ sim/ tb_top/ testcases/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../testcases/
ahb_sramc_test_base.sv ahb_sramc_test_pkg.sv seqlib/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../
agent/ assertion/ coverage/ env/ lists/ model/ ral/ sim/ tb_top/ testcases/
<verwcj-~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts> $g ../../tb_top/ahb_sramc_tb.sv
1:12:46 / 1:32:48
护眼: 关
x2倍速
高清


tb.f (~/keller_work/test_ing/sramc_uvmtb_unsync/ahb_sramc/verify/sim/scripts) - GVIM3
File Edit Tools Syntax Buffers Window UVM Verilog Help
1
2 incdir+../../agent/ahb
3 ./../agent/ahb/ahb_pkg.sv
4 incdir+../../env
5 ./../env/ahb_sramc_env_pkg.sv
6 incdir+../../testcases
7 ./../testcases/ahb_sramc_test_pkg.sv
8 incdir+../../tb_top
9 ./../tb_top/ahb_sramc_tb.sv
10
www.coreshinelearning.com userhd7y06
"tb.f" [New] 10L, 214C written
-- VISUAL --
3,1 All
1:13:11 / 1:32:48
护眼: 关
x2倍速
高清